

PHYSICS GRADE 10: ENERGY, WORK, POWER AND MACHINES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.4 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 1.0: Mechanics and Thermal Physics
Sub-Strand	Sub-Strand 1.4: Energy, Work, Power and Machines
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Forms of energy: kinetic energy and potential energy (gravitational and elastic) – Energy transformation and conservation of energy – Work: definition, calculation, positive and negative work ($W = Fd$) – Work done against friction – Power: definition, calculation and everyday applications ($P = W/t$) – Simple machines: levers, pulleys, inclined planes (ramps) – Mechanical advantage, velocity ratio and efficiency of machines – Elastic potential energy in springs (Hooke's Law) – Energy conversion in electrical devices such as fans and motors
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - distinguish between kinetic energy and potential energy and calculate each in everyday situations, - apply the principle of conservation of energy to explain energy transformations such as in a roller coaster or pendulum, - define work and calculate work done by a force, including positive and negative work and work done against friction, - define power and calculate it for everyday activities such as climbing stairs or lifting water, - explain how simple machines (pulleys, levers, ramps) reduce effort, and determine their mechanical advantage, velocity ratio and efficiency, - appreciate the application of energy, work, power and machines in day-to-day life.
Core Competencies	– Critical thinking and problem solving: learners analyse data and reason about why machines reduce effort without reducing the work done, and solve numerical problems on work, energy and power. – Communication and collaboration: learners work in groups during experiments and share evidence-based explanations of energy transfers with the class. – Creativity and imagination: learners design and test a simple machine, such as a pulley or ramp system, for a real Kenyan task like lifting water or loading sacks of maize. – Digital literacy: learners use PhET simulations (Energy Skate Park, Pendulum Lab, Forces and Motion) to investigate energy and work. – Learning to learn: learners revise their models and track their growing understanding using the Driving Question Board.

Core Values	– Responsibility: learners handle apparatus safely and take on assigned roles during group investigations. – Integrity: learners record and report experimental measurements honestly, even when results differ from expectations. – Unity: learners cooperate in teams to design, test and improve simple machines.
Science & Engineering Practices	– Asking questions and defining problems: learners generate questions about why a pulley or ramp makes work easier and add them to the Driving Question Board. – Developing and using models: learners build and revise an energy-flow model from the well, ramp, lever and roller coaster scenarios. – Planning and carrying out investigations: learners measure effort, load and distance for pulleys and ramps. – Analyzing and interpreting data: learners compare input and output work to calculate efficiency. – Using mathematics and computational thinking: learners apply $W = Fd$, $P = W/t$, $KE = \frac{1}{2}mv^2$ and $PE = mgh$. – Constructing explanations: learners use evidence to explain how energy is conserved and transferred.
Pertinent & Contemporary Issues (PCIs)	– Environmental education: learners relate energy efficiency to saving fuel and electricity and reducing environmental impact. – Safety and security: learners practise safe lifting and use of tools such as crowbars, scissors and pulleys. – Education for sustainable development: learners explore energy-saving machines and appropriate technology for rural communities.
Career Connections	– Mechanical engineer: designs machines and improves their efficiency. – Civil and construction engineer: uses pulleys, cranes and ramps to lift and move heavy loads. – Agricultural engineer: designs water-lifting systems, irrigation pumps and farm tools. – Automotive technician (Jua kali mechanic): repairs engines, levers and gear systems. – Electrical engineer: designs motors and fans that convert electrical energy to mechanical work. – Sports scientist: analyses the work and power of Kenyan athletes and cyclists.
Focus for Lessons	This unit uses everyday Kenyan experiences, such as lifting water from a well with a pulley, to build understanding of energy, work, power and machines. Learners predict, investigate with hands-on activities and simulations, explain using evidence and mathematics, and continuously revise a model that shows how machines change the effort needed while energy is conserved.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process? KICD KEY INQUIRY QUESTIONS: 1. How do we use energy in our day-to-day life? 2. How do machines make work easier?
Anchoring Phenomenon	At a rural homestead in Kitui County, a family draws water from a deep well. When a woman lifts a full 20 kg bucket of water straight up by hand, it is exhausting and she can barely raise it. When the same bucket is hooked onto a rope passing over a pulley fixed above the well, she lifts it with much less effort, and a child can even do it. Yet the bucket still rises the same height and gains the same amount of energy. Learners are puzzled: if the bucket gains the same energy, why does it feel so much easier with the pulley, where does the energy come from, and is any energy lost along the way? Using a pulley system makes lifting heavy buckets easier.
Storyline Thread	Lesson 1: Kinetic and potential energy. Learners predict how a bucket's energy changes as it is lifted from the well, then use PhET Energy Skate Park to observe kinetic and potential energy, and begin the DQB and initial model. Lesson 2: Energy transformations. Learners investigate a pendulum and roller coaster, calculate $KE = \frac{1}{2}mv^2$ and $PE = mgh$, and add transformations to their model.

	Lesson 3: Conservation of energy. Using PhET Pendulum Lab and a demonstration, learners test whether total energy stays constant and revise the model with energy conservation. Lesson 4: Conservation of energy in applications. Learners solve problems on falling and sliding objects and account for energy losses as heat and sound, linking back to the bucket in the well. Lesson 5: Work. Learners define work ($W = Fd$), calculate work done lifting a bucket, and distinguish positive and negative work using everyday examples. Lesson 6: Work and energy. Learners connect work done to change in energy, compare work done by hand and with a pulley, and refine the DQB with new questions. Lesson 7: Work done against friction and power. Learners investigate dragging a sack of maize on rough and smooth surfaces, and define and calculate power ($P = W/t$) using stair-climbing and water-lifting examples. Lesson 8: Simple machines and efficiency. Learners test pulleys, levers and ramps, calculate mechanical advantage, velocity ratio and efficiency, and complete their final model to answer the driving question on how pulleys reduce effort while energy is conserved.
Supporting Phenomena	– Opening a door with a handle: a door moves easily when pushed at the handle but is much harder to open near the hinge. – Using a ramp to load heavy objects: rolling a box (or a sack of maize) up a ramp onto a lorry in Nairobi requires less effort than lifting it vertically. – Stretching a spring (Hooke's Law): stretching a spring stores energy that can do work when released. – Cycling uphill vs. flat road: a boda boda rider or cyclist in the Rift Valley works much harder going uphill than on flat ground. – Using scissors or a crowbar: cutting or prying requires less effort with simple tools. – Energy transformation in a roller coaster: a coaster converts potential energy at the top into kinetic energy on the track. – Electric fan or motor work: electrical energy is converted into mechanical work to rotate the fan.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (40 minutes (one lesson; the five phases are timed within it)): Kinetic and Potential Energy: The Bucket in the Well

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the Kitui well phenomenon, introduce energy as something stored by position (potential) or present because of motion (kinetic), and open the Driving Question Board and an initial qualitative model.
Knowledge	Learners will be able to: (a) state that potential energy is energy stored because of position and kinetic energy is energy an object has because it is moving; (b) explain, in words, that a bucket lifted higher in the well stores more potential energy (the PE table shows a rise from 0 J at the water to 1600 J at ground level); (c) describe from PhET Energy Skate Park that potential and kinetic energy exchange as a skater moves up and down.
Skills	Predicting, observing and recording evidence from a simulation, identifying patterns (higher position means more stored energy), asking investigable questions, drawing a first qualitative model (NGSS Science and Engineering Practices: Developing and Using Models, Asking Questions, Analysing Data, Engaging in Argument from Evidence).
Attitudes	Curiosity about how everyday water-drawing technology works; respect for peers' ideas, including incomplete ones; responsibility in contributing to group work; appreciation of how Physics explains the work of families such as Mama Kavata's in Kitui.
Key Inquiry Question	1. How do we use energy in our day-to-day life? 2. How do machines make work easier?
Purpose in Storyline	Lesson 1 launches the phenomenon. Learners meet Mama Kavata Mutua, her daughter Mwende and the pulley system in the 8 m deep well in Kitui, and gather the first ideas (potential and kinetic energy) needed for later lessons. Their initial predictions, questions and model are revisited and revised in every later lesson; no formulas are taught yet.
Safety Notes	Computer/tablet use: seat learners in groups of 3-4 with one device each, keep cables off walkways. If doing the optional hands-on lift, use a light bag (about 1 kg) or a 1 L bottle on a short string at table height; never lift loads over heads or feet. No learner should lean over any real well; remind them to stay away from open wells at home and always have an adult present.

B. LESSON OVERVIEW

Mr Otieno opens the class with a short story from Kitui County: Mama Kavata Mutua lifts a full 20 kg bucket (weight 200 N) straight up from an 8 m deep well and is exhausted, yet her 10-year-old daughter Mwende lifts the same bucket easily using the pulley system. The bucket rises the same 8 m either way. Learners predict how the bucket's energy changes as it rises, then use the PhET Energy Skate Park simulation to observe potential and kinetic energy exchanging as the skater moves. They apply the observations back to the bucket (the PE trend from 0 J at the water to 1600 J at the top, qualitative only), open the Driving Question Board with their own questions, and sketch an initial model of the bucket, the person and energy. The lesson ends with learners naming what they observed, learned and could explain so far, leaving the key puzzle open: why does the pulley make lifting feel easier, where does the energy come from, and is any lost?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	(8 min) Learners listen to or read the Kitui story: Mama Kavata lifts	Tell the story slowly, showing a photo or drawing of a Kitui well	Predict-Observe-Explain (POE) with a Claim-and-Reason prompt:	Check prediction papers for: (1) a clear claim about whether the

 VIDEO: Introduction to energy Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Energy Skate Park: Basics Source: PhET Interactive Simulations (English)  Search ARES for similar readings	the 20 kg bucket 8 m by hand and struggles, while Mwende lifts it using the pulley system. They sketch the well on a half page of paper (water at the bottom, ground at the top, with the bucket at 0 m, 2 m, 4 m, 6 m and 8 m marked) and answer individually: 'As the bucket is lifted from the water to the top, what happens to its energy? Does it gain, lose or keep the same energy? Will the energy gained be different for Mama Kavata and for Mwende?' They write a prediction with a reason in 2-3 sentences, then share with a partner and agree or disagree.	and pulley. Ask: "Have you or your family ever drawn water from a well or borehole? What did it feel like?" Take 3 quick responses. Then say: "Before I teach anything, I want your best thinking. There are no wrong predictions today." Pose the prediction question and give 2 minutes of silent writing. Then WAIT TIME 10-15 seconds after asking: "Who thinks the bucket's energy changes as it goes up? Show me with a hand." Cold-call 4 learners (2 who predict it increases and 2 who predict it stays the same or decreases): "Naomi, what is your reason?" Do not correct. Record predictions in three columns on the board (Gains energy / Same / Loses energy) with tallies. Say: "Keep your prediction paper; we will check it at the end."	committing to a prediction with a reason exposes learners' existing ideas about energy (for example, 'energy is the same as force' or 'the pulley gives energy') so that later evidence can be compared against them and conflicts become productive.	bucket's energy increases, decreases or stays the same; (2) a reason that mentions height, position or effort; (3) whether learners distinguish 'energy' from 'effort/force'. Tally on the board how many predicted 'gains energy'. Note any who say Mwende's bucket gains less energy because it felt easier; this is the key misconception to track.
Observe Phase  VIDEO: Work and energy (part 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Energy Skate Park: Basics Source: PhET Interactive Simulations (English)  Search ARES for similar readings	(12 min) In groups of 3-4, learners open PhET Energy Skate Park (offline copy, 'Intro' screen) and select the bar graph and the pie chart displays. Roles: operator, recorder, reader, reporter. Task 1: Release the skater from the highest point on the track and watch the energy bars. Task 2: Pause at the top, at the middle and at the bottom of the track and sketch or describe the bars (potential, kinetic, thermal/total) in a 3-row table: Position / Potential energy (high-medium-low) / Kinetic energy (high-medium-low) / Speed (fast, slow, still). Task 3: Start the skater from a lower point and compare. They answer: 'When is the skater's potential energy greatest? When is the kinetic energy greatest? What happens to the total energy bar?' (Friction is set to none for the first run; they	Demonstrate how to open the simulation in 1 minute, then circulate. Ask each group: "Where is the skater when the potential energy bar is tallest? How do you know the skater has kinetic energy there or not?" Prompt: "What is the skater doing when the kinetic bar is tallest?" Give WAIT TIME of 10-15 seconds before accepting answers. Use the talk-move 'Say more about that.' For groups that finish, ask: "If the skater started lower, what would happen to the highest potential energy bar?" Then pull the class together: show the bucket PE table on the board (heights 0, 2, 4, 6, 8 m; energy 0, 400, 800, 1200, 1600 J) and ask: "What pattern do you see as the bucket goes up?" Cold-call 3 learners. Introduce the vocabulary only after learners have described the ideas in their own words:	Structured observation with a data table and pattern-finding: recording energy bars at set positions turns a moving simulation into comparable evidence. The stage-by-stage 'describe first, name later' vocabulary approach lets learners build meaning for potential and kinetic energy from what they see before receiving the terms. Transferring the pattern to the bucket PE table links the simulation to the phenomenon.	Observe whether groups: (1) correctly identify the top as highest PE and zero or lowest KE; (2) identify the bottom as highest KE and lowest PE; (3) record that PE and KE trade off while the total bar stays about the same with no friction; (4) state the bucket trend as 'higher means more potential energy'. Use a quick thumb check: 'Thumbs up if you can say where the skater's kinetic energy is greatest.' Note groups who confuse speed with height.

	may turn it on in the last 2 minutes to notice thermal energy appearing, without explaining it yet.) Then learners link to the bucket: using the bucket PE table (0 J at the water, 400 J, 800 J, 1200 J, and 1600 J at ground level) they describe the trend in words: 'The higher the bucket, the ____ its potential energy.'	"Energy stored because of position is called potential energy. Energy because of motion is called kinetic energy." Do not introduce $PE = mgh$ or KE formulas; those come in Lesson 2. If a learner asks how the numbers were found, say: "Good question. Put it on our board; we will learn how to calculate it next lesson."		
Explain Phase  VIDEO: Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Average Kinetic Energy (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	(8 min) Learners return to the Kitui story and individually write a short explanation (3-4 sentences) answering: 'Using what you saw with the skater, explain what is happening to the bucket's energy as Mama Kavata lifts it from the water to the top of the well. What kind of energy is it gaining? What would happen to that energy if the rope slipped and the bucket fell back?' They then do a 'Think-Pair-Share' and compare with a partner. Finally, the class discusses: 'Mama Kavata and Mwendu raise the bucket the same 8 m. Does the bucket have the same potential energy at the top in both cases? Why might it still feel so different to lift?' Learners are encouraged to offer ideas about the pulley without being told the answer.	Say: "Use the words potential and kinetic in your explanation. Where does the bucket have the most potential energy? What happens to that stored energy when it falls?" Give 3 minutes of silent writing, then WAIT TIME 10-15 seconds before pair sharing. Cold-call 4 learners to read their explanations, calling on at least one who used the skater as evidence: "Brian, which part of the simulation helped you say that?" Use the talk move: "Who can add to or challenge that idea?" Then ask the bigger question and take 3 ideas without judging: "The bucket ends up with the same stored energy either way, yet Mwendu finds it easier. What might the pulley be doing?" Record unanswered parts on the board as 'We still wonder...'. Say: "Today we have one piece of the puzzle: the bucket stores more potential energy the higher it goes. We do not yet know how the pulley helps." Do not explain mechanical advantage or work yet.	Evidence-based explanation (Claim-Evidence-Reasoning, simplified) with Think-Pair-Share: learners must connect a claim about the bucket to evidence from the skater simulation and the PE trend. Naming what remains unexplained keeps the cognitive conflict about the pulley open and creates a purpose for later lessons.	Look in written explanations for: (1) 'potential energy' used correctly for the bucket at height; (2) a statement that higher position means more stored energy; (3) recognition that falling would turn potential energy into kinetic energy (moving energy); (4) no claim that the pulley adds energy. Record names of learners whose explanation says the pulley gives energy or that the bucket has less energy with the pulley, for revisiting in Lesson 6.
Driving Question Board (DQB) Creation  VIDEO: Introduction to energy Source: Khan	(7 min) Learners individually write 2 questions on sticky notes or paper strips about the Kitui bucket story, such as 'Why does the pulley make it easier?', 'Where does the energy to lift the bucket come from?', or 'Is any energy lost?'. In groups they sort the	Post the Driving Question on a chart in the centre of the board. Say: "What do you wonder about Mama Kavata's well? Write questions we can investigate, not only yes or no questions." Show a model: 'Where does the energy come from when Mwendu pulls?'	Driving Question Board: learner-generated questions give ownership of the phenomenon and a visible record of what the class needs to figure out. Sorting questions into categories reveals the big ideas (energy, force, machines, loss) the unit will	Count how many questions refer to energy transfer or the pulley mechanism rather than only to descriptions. Look for at least one question on each of: where energy comes from, why the pulley is easier, and whether energy is lost. Note learners whose questions

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Average Kinetic Energy Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	questions into categories (energy, force and effort, machines, losses) and place them on the class DQB around the central Driving Question: 'How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process?' Each group reads out one question and explains why it matters. Learners also add two KICD inquiry questions: 'How do we use energy in our day-to-day life?' and 'How do machines make work easier?'	Give 3 minutes of writing time. Then WAIT TIME 10-15 seconds, and cold-call 5 groups to place and read one question each. Ask: "Does another group have a similar question? Let's put them together." Highlight questions that link to ideas to come: food energy, friction, work, machines. Say: "We will come back to this board every lesson and move questions to the 'answered' side when we have evidence." Take a photograph of or copy the board so it can be recreated on the offline platform.	address and allows learners to track their growing understanding.	show the misconception that the pulley creates energy; this is useful for later discussion. Check that groups can justify where their question belongs on the board.
Model Building Phase VIDEO: Work and energy (part 2) Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Thermal & Kinetic Energy (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	(5 min) Individually, then in groups, learners draw an initial model of the Kitui well on a sheet of paper or mini whiteboard. They draw the well (water at 0 m, ground at 8 m), the 20 kg bucket at two positions, and the person (Mama Kavata, and Mwende with the pulley system). They add labels and arrows showing: where the bucket stores potential energy, where they think the energy to lift it comes from (for example food such as ugali or githeri), and any questions about what the pulley does. No formulas are used. Beneath it they complete a mini summary: 'We observed... We learned... We can explain... We still wonder...'	Say: "A model is our best current picture of how something works. It will change as we learn more. Show the bucket at the bottom and the top, and show energy with words, arrows or colours." Circulate and ask: "What does this arrow show? Where does the energy come from? What does the pulley do in your model?" WAIT TIME 10-15 seconds before responding. Cold-call 3 groups to hold up and describe their model in 30 seconds each. Remind them that a model is not wrong because it is incomplete: "Leave empty spaces or question marks where you are not sure." Collect or photograph models and say: "We will return to this model in Lesson 2 and revise it when we learn how to calculate energy." Finish by revisiting the prediction board: "Look back at your prediction. Did the bucket's energy increase as it rose? What evidence did you see?"	Initial model building (qualitative): drawing a model forces learners to represent relationships between height, stored energy and the source of energy in one picture, revealing gaps and misconceptions. Treating the model as a draft to be revised builds the habit of cumulative model revision across the unit.	Check models for: (1) potential energy shown as higher at the top of the well than at the water; (2) an energy source for the person (food) or an unlabelled question mark; (3) a bucket that gains the same energy for both Mama Kavata and Mwende, or a clear question about it; (4) labels using 'potential' or 'kinetic' correctly. Use a quick exit prompt: 'One thing I now know about the bucket's energy is ____.' Review the exit responses to plan Lesson 2.

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners predict that the bucket gains energy as it rises, and could they give reasons beyond 'it is heavy'? (2) Did learners separate energy stored by position from energy of motion, or did they mix up height and speed? (3) Which misconceptions appeared (for example the pulley gives extra energy, or the bucket has less energy with the pulley) and how many learners held them? (4) Which DQB questions are most ready to be investigated in the next lessons, and which will need to wait until Lesson 6 or 8? (5) Did learners with limited device access take part fully, and should groups or roles be adjusted? (6) Were the PhET tasks and the bucket PE table clearly connected so that learners could see the skater and the bucket as the same energy idea? Note the timing of each phase and adjust Lesson 2 accordingly.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the Energy Skate Park simulation we observed that the skater has the greatest potential energy at the highest point and the greatest kinetic energy at the lowest point, with energy moving between the two as the skater moves. We also observed from the bucket table that the higher the bucket is in the Kitui well, the more potential energy it has, rising from 0 J at the water to 1600 J at the top.
What did I learn?	Potential energy is energy stored because of position, such as a bucket lifted up the well, and kinetic energy is energy an object has because it is moving. A bucket lifted higher stores more potential energy, and stored energy can change into movement energy when something falls.
How does this explain the phenomenon?	We can explain that Mama Kavata and Mwendu both raise the bucket the same 8 m, so the bucket gains stored (potential) energy as it rises in both cases. We cannot yet explain why the pulley makes it easier, where the energy for lifting comes from, or whether any energy is lost; these questions are on our Driving Question Board for the lessons ahead.

LESSON 2 (80 minutes (2 x 40-minute periods)): Energy Transformations and the Formulas $KE = \frac{1}{2}mv^2$ and $PE = mgh$

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners calculate gravitational potential energy ($PE = mgh$) and kinetic energy ($KE = \frac{1}{2}mv^2$) and trace how energy changes between PE and KE in a pendulum and a roller coaster. They then use the same ideas to explain how the energy of the 20 kg bucket in the Kitui well changes with height.
Knowledge	Learners will be able to: (a) state and use $PE = mgh$ with $g = 10 \text{ N/kg}$, with height measured from a chosen zero level (the water surface at 0 m); (b) state and use $KE = \frac{1}{2}mv^2$; (c) calculate PE and KE in joules for the pendulum bob (0.5 kg, raised 0.2 m, 1 J at the top, 2 m/s and 1 J at the bottom), the roller coaster car (1000 kg, 20 m high, 200 000 J, 20 m/s at the bottom) and the bucket (400 J, 800 J, 1200 J, 1600 J at 2, 4, 6, 8 m); (d) describe energy transformations as PE to KE on the way down and KE to PE on the way up.
Skills	NGSS Science and Engineering Practices: Using Mathematics and Computational Thinking (substitute values into formulas and check units); Analyzing and Interpreting Data (read PE and KE columns in tables and graphs); Developing and Using Models (add energy transformation arrows to the class model); Constructing Explanations (use evidence from the pendulum and roller coaster to explain the bucket). Core competencies: Critical thinking and problem solving; Communication and collaboration.
Attitudes	Values: Responsibility (each group member takes a role: reader, calculator, recorder, reporter); Respect (listening to the reasoning of peers when predictions differ). Learners appreciate that mathematics lets us predict energy in everyday situations such as drawing water in Kitui.
Key Inquiry Question	How do we use energy in our day-to-day life? How do machines make work easier? (This lesson focuses on the first question by tracing where energy is stored and how it changes form.)
Purpose in Storyline	Lesson 1 gave learners the ideas of potential and kinetic energy and a qualitative initial model of the bucket in the well. Lesson 2 makes those ideas quantitative with $KE = \frac{1}{2}mv^2$ and $PE = mgh$, and shows that energy changes form between PE and KE. It prepares Lesson 3, where learners test whether the total of $KE + PE$ stays constant. Learners leave knowing the bucket gains 1600 J of PE when raised 8 m, which is the number the driving question must explain. Other topics are left to later lessons.
Safety Notes	Pendulum activity: use small bobs (a stone or 50 g mass on string) swinging away from faces; keep the swing area clear and stand clear of other groups. Tie strings to a firm support (desk edge clamped or classroom beam). Do not stand under a raised object. Roller coaster is watched by video only. Keep floors dry and bags out of the aisles.

B. LESSON OVERVIEW

Mr Otieno opens by reminding learners of Mama Kavata's bucket in Kitui: the 20 kg bucket gains the same energy whether she lifts it by hand or with the pulley system. Lesson 1 taught that a raised bucket stores potential energy and a moving one has kinetic energy. Today learners ask: how much energy, and how does it change form? They first predict what happens to a swinging pendulum and a roller coaster car, then observe a real pendulum and a roller coaster video. They derive and apply $PE = mgh$ and $KE = \frac{1}{2}mv^2$ using the pendulum (0.5 kg, 0.2 m, 1 J, 2 m/s), the roller coaster (1000 kg, 20 m, 200 000 J, 20 m/s) and the bucket PE table (400 J at 2 m up to 1600 J at 8 m). In the Explain phase learners decide what the numbers say about the bucket, then add energy transformation arrows and formulas to the DQB and class model. Learners finish knowing that the bucket gains 1600 J whichever way it is lifted, but not yet why the pulley feels easier; that stays on the DQB.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Conservation of energy Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Energy Skate Park: Basics Source: PhET Interactive Simulations (English)  Search ARES for similar readings	(10 min) Learners recall the Kitui bucket and the Lesson 1 idea that height stores potential energy. Mr Otieno holds a 0.5 kg bob on a string at a small height and asks learners to predict individually in their notebooks: (1) What happens to the bob's speed as it swings down? (2) Where is it fastest? (3) Do you think a heavier bob or a higher release stores more energy? Then show a picture of a roller coaster car at the top of a 20 m hill and predict where it moves fastest. Learners sketch a quick graph of 'PE and KE against position' for the pendulum and compare with a partner.	Say: 'Remember Mama Kavata's bucket. Lesson 1 told us that height stores energy. Today we ask how much, and what happens to it when things move.' Hold the bob at the top and ask: 'Before I let go, write down what you think happens to its speed and its energy.' WAIT TIME 10-15 seconds in silence while learners write. Then cold-call 4 learners from different groups, asking each: 'What did you predict and what is your reason?' Record 2-3 predictions on the board without judging. Ask: 'Does the roller coaster car follow the same pattern as the bob?' Wait 10 seconds before taking answers. Note any belief that energy disappears at the bottom.	Predict-Observe-Explain (POE) with a quick-write: committing to a prediction in writing makes learners' existing ideas visible and gives them a reason to check those ideas against evidence. Pairing the pendulum with the roller coaster lets learners test whether one idea explains two different situations.	Observable indicators: (a) learners write a prediction and a reason, not just a prediction; (b) at least 3 of 4 cold-called learners link height to stored energy using Lesson 1 vocabulary (potential, kinetic); (c) note on a tally who predicts that energy 'runs out' at the bottom, to revisit in Lesson 3.
Observe Phase  VIDEO: Conservation of energy Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Energy Skate Park: Basics Source: PhET Interactive Simulations (English)  Search ARES for similar readings	(25 min) Part A, pendulum (groups of 4, 12 min): Each group hangs a bob from a firm support, raises it 0.2 m above its lowest point, releases it and watches. Learners record where the bob is fastest and slowest and where it momentarily stops. Mr Otieno then gives the measured data for the 0.5 kg bob: raised 0.2 m, speed at the bottom 2 m/s. Learners complete a table: PE at top = $mgh = 0.5 \times 10 \times 0.2 = 1 \text{ J}$; KE at the bottom = $\frac{1}{2}mv^2 = \frac{1}{2} \times 0.5 \times 2^2 = 1 \text{ J}$. Part B, roller coaster (13 min): Learners watch a short roller coaster video (for example at a theme park in Nairobi or similar) and a diagram of a 1000 kg car at the top of a 20 m hill. In groups they calculate PE at the top = $1000 \times 10 \times 20 = 200\,000 \text{ J}$ and KE at	Circulate during Part A and ask: 'Where is the bob moving fastest? Where does it stop for an instant?' Prompt: 'Which position has the most height? What does that tell you about its stored energy?' Give a role card to each group member (reader, calculator, recorder, reporter). Before the calculations, show the formulas $PE = mgh$ and $KE = \frac{1}{2}mv^2$, with m in kg, g = 10 N/kg, h in m and v in m/s, and show the unit check: $\text{kg} \times \text{N/kg} \times \text{m} = \text{N} \times \text{m} = \text{J}$. Model the first calculation aloud on the board, writing every substitution, then let groups do the rest. WAIT TIME 10-15 seconds after asking: 'Compare your PE at the top with your KE at the bottom. What do you notice?' Cold-call 3 groups for their two numbers (1 J and 1 J; 200 000 J	Data table with a pattern-finding routine: learners fill in structured tables so the numbers, not the teacher, show that PE at the top and KE at the bottom match. Using a hands-on pendulum first and a large-scale roller coaster second builds a bridge from small, safe observations to a bigger real-life case, so the pattern is seen in two different contexts.	Observable indicators: (a) learners substitute correctly into $PE = mgh$ and $KE = \frac{1}{2}mv^2$ and write answers in J (check at least 1 J, 1 J, 200 000 J, 200 000 J); (b) common errors to catch: forgetting to square v (writing $\frac{1}{2} \times 0.5 \times 2 = 0.5 \text{ J}$), using g = 9.8 instead of 10, or leaving out units; (c) learners point to the lowest point of the swing as the place of the highest speed and the highest point as the place where the bob stops; (d) the recorder's table matches the data given.

	the bottom with $v = 20 \text{ m/s}$: $\frac{1}{2} \times 1000 \times 20^2 = 200\,000 \text{ J}$ (ideal, no friction). They mark on a diagram where PE is largest, where KE is largest, and where the car is part-way.	and 200 000 J). Do not yet name conservation of energy: say 'Hold that thought, we will test it properly in Lesson 3.'		
Explain Phase  VIDEO: Conservation of energy - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Average Kinetic Energy (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	(25 min) Part A (10 min): Learners construct a PE and KE graph for the free-fall of the bucket using the table (height 8, 6, 4, 2, 0 m with PE 1600, 1200, 800, 400, 0 J and KE 0, 400, 800, 1200, 1600 J). First they use $PE = mgh$ with $m = 20 \text{ kg}$ to confirm the PE column (for example at 2 m: $20 \times 10 \times 2 = 400 \text{ J}$). They note that PE falls as KE rises. Part B (10 min): In a Claim-Evidence-Reasoning (CER) paragraph, each learner explains: 'When the bucket falls from the top of the well to the water, what happens to its energy? Use numbers from the table, the pendulum and the roller coaster.' Part C (5 min): Using the bucket PE table, learners answer: 'When Mama Kavata lifts the bucket from 0 m to 8 m, how much PE does the bucket gain? Does this change if Mwende lifts it using the pulley system?' Learners discuss in groups that the PE gain depends only on the weight (200 N) and the height gained (8 m): $200 \times 8 = 1600 \text{ J}$, for either method.	Ask: 'Look at the PE column, then the KE column. What happens to one when the other goes up?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Press for evidence: 'Which number in the table shows that?' Provide the CER sentence frames: 'I claim that... My evidence is... This shows that...'. Ask: 'In the pendulum and the roller coaster, energy changed form. Name the form at the start and the form at the bottom.' (PE to KE, and KE back to PE on the way up.) For Part C ask: 'Mama Kavata lifts by hand; Mwende uses the pulley system. The bucket rises the same 8 m. What is the PE gain for each?' Cold-call 2 learners, then ask a third to restate the first answer in her own words. Then say: 'So the PE gain is 1600 J either way. Your question now is the one on our board: if the energy gain is the same, why does the pulley feel easier? We cannot answer that yet. Which lessons will help?' Record learner ideas without correcting them; park them on the DQB. Common misconception to address: 'a pulley gives the bucket less energy'. Correct only with evidence: 'Check $PE = mgh$. Is there any place for a pulley in that formula?'	Claim-Evidence-Reasoning (CER) with graph reading: learners must back each claim about energy transformation with numbers from the tables. Transferring the pattern from the pendulum and roller coaster to the bucket requires them to apply the formulas to a new case, which shows what stays the same (the 1600 J gain) and what remains unexplained (the smaller effort), keeping the driving question open.	Observable indicators: (a) CER paragraphs state that PE decreases as KE increases using at least two numbers from the table (for example 1600 J PE and 0 J KE at 8 m, 0 J PE and 1600 J KE at 0 m); (b) learners correctly compute that the bucket gains 1600 J at 8 m whichever method is used; (c) learners use 'transformed' or 'changes from... to...' rather than 'used up'; (d) note who still thinks the pulley changes the bucket's PE and give targeted follow-up.
Driving Question Board (DQB) Creation  VIDEO:	(10 min) Learners return to the Driving Question Board started in Lesson 1. On sticky notes they (1) write one new question that arose from today's investigations (for	Say: 'Look at the board. Which of our questions did we answer with numbers today?' WAIT TIME 10-15 seconds before taking responses. Ask 3 learners to place	Driving Question Board with question sorting: sorting learners' own questions makes visible what the evidence has and has not explained, gives learners	Observable indicators: (a) each learner adds at least one new question that refers to today's evidence (energy, height, speed, formulas) rather than a general

Work and energy (part 2) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Average Kinetic Energy Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	example 'Is the total KE + PE always the same?', 'Where does the energy come from when Mwende pulls the rope?', 'Does friction change the numbers?'), and (2) move or mark any earlier questions now partly answered (for example 'How much energy does the bucket gain?' is answered: 1600 J). They sort the questions into 'answered today', 'partly answered' and 'still to investigate'.	a note and say why. Point to the unresolved question about why the pulley feels easier and say: 'We know the energy gain. What is missing is how the person supplies the energy and why the effort changes.' Invite 2 learners to read aloud a new question. Group questions under headings: 'Does total energy stay the same? (Lesson 3)', 'What happens when there is friction? (Lessons 4 and 6)', 'How does force and distance relate to the effort? (Lessons 5, 6, 8)'. Do not answer these now.	ownership of the unit's direction and links each future lesson to a real need to know.	question; (b) at least two questions point to conservation or energy loss, showing readiness for Lessons 3-4; (c) learners can say which board question was answered today and quote the number (1600 J).
Model Building Phase VIDEO: Video: Energy Skate Park: Basics <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar videos HTML: Average Kinetic Energy (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	(10 min) Learners return to the initial model of the bucket in the well from Lesson 1. In groups they revise it: label the bucket at 0, 2, 4, 6 and 8 m with the PE values (0, 400, 800, 1200, 1600 J); add the formula $PE = mgh$ with the bucket's numbers; add arrows showing 'PE to KE' when the bucket falls and 'KE to PE' on a pendulum's upward swing (a small inset of the pendulum with 1 J and 2 m/s); add the chemical energy in food (ugali, githeri) as the source of the energy Mama Kavata or Mwende uses to raise the bucket, as a question mark to test later. Each group writes one sentence at the bottom: 'The bucket gains ____ J of PE when raised 8 m, by hand or with the pulley system.'	Say: 'A model is a tool to explain and predict. What new parts does our model need after today?' Hand out the Lesson 1 models. WAIT TIME 10-15 seconds while groups decide what to add. Visit each group and ask: 'Where on your model can I see the 1600 J?' and 'What does this arrow mean in joules?' Cold-call 3 groups to share one revision and the evidence for it. Make sure every model shows the formulas, the 400 to 1600 J values and PE-KE arrows. Display the models on the wall. Close with: 'Our model now tells us how much energy the bucket gains. Next lesson we ask whether the total energy changes during the fall.'	Cumulative model revision: revising the same model each lesson shows learners how their understanding grows and forces them to turn numerical evidence into a visual explanation. Labelling with numbers and arrows makes it testable.	Observable indicators: (a) the model includes all five PE values for the bucket with correct units; (b) $PE = mgh$ and $KE = \frac{1}{2}mv^2$ are written with the meaning of each symbol; (c) arrows show PE to KE and KE to PE correctly; (d) the sentence states 1600 J and says it applies to both lifting methods; (e) groups can say what is still not in the model (why the pulley needs less effort).

D. TEACHER REFLECTION

After the lesson ask yourself: (1) Could learners calculate KE and PE correctly with units, and which errors were most common (forgetting to square v, using 9.8 instead of 10, confusing mass and weight)? (2) Did learners see the PE of the bucket as depending on weight and height only, not on how it is lifted? Who still thinks the pulley changes the PE? (3) Did any learners say that energy 'disappears' at the bottom of the swing? Plan to revisit this in Lesson 3. (4) Were all learners participating in the group roles, and did the 10-15 second wait time lead to more thoughtful answers? (5) Which learners need extra practice with substitution before Lesson 3, where the formulas will be used again? Note which DQB questions can guide the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We swung a 0.5 kg pendulum bob raised 0.2 m and watched it move fastest at the bottom and stop at the top of its swing. We saw a roller coaster car speed up going down a hill and slow down going up. We read the free-fall table for the 20 kg bucket from 8 m to 0 m, where PE falls from 1600 J to 0 J as KE rises from 0 J to 1600 J.
What did I learn?	PE = mgh and KE = $\frac{1}{2}mv^2$, with g = 10 N/kg and energy in joules. The pendulum has 1 J of PE at the top and 1 J of KE (2 m/s) at the bottom. The roller coaster car has 200 000 J of PE at the top and 200 000 J of KE (20 m/s) at the bottom in the ideal case. Energy changes form between PE and KE. The bucket gains 400, 800, 1200 and 1600 J of PE at 2, 4, 6 and 8 m.
How does this explain the phenomenon?	The bucket in the Kitui well gains 1600 J of PE when raised 8 m (200 N x 8 m), whether Mama Kavata lifts it by hand or Mwende uses the pulley system, because PE depends only on weight and height. This does not yet explain why the pulley feels easier or where the energy comes from; those questions stay on the DQB for the next lessons.

LESSON 3 (80 minutes (2 x 40-minute periods)): Conservation of Energy: Does the Total Stay the Same? Pendulum and the Falling Bucket

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners test whether the total energy (KE + PE) of a pendulum and of a freely falling bucket stays constant in an ideal system, and use this principle to revise the class model of the Kitui bucket.
Knowledge	In an ideal system with no friction, KE + PE stays constant: energy is never created or destroyed, only transformed or transferred. For the bucket released from 8 m, KE + PE = 1600 J at every height. For the pendulum bob (0.5 kg raised 0.2 m), total energy is 1 J throughout.
Skills	Predicting, using a simulation (PhET Pendulum Lab) to collect data, calculating KE and PE with $KE = 1/2 mv^2$ and $PE = mgh$ ($g = 10 \text{ N/kg}$), completing and interpreting data tables, constructing and revising a model, arguing from evidence.
Attitudes	Curiosity about where energy goes, respect for evidence and for others' ideas during group talk, and responsibility in sharing roles. Appreciation that the same energy rules govern a village well in Kitui and machines everywhere.
Key Inquiry Question	1. How do we use energy in our day-to-day life? 2. How do machines make work easier?
Purpose in Storyline	Lesson 2 taught learners to calculate KE and PE and trace how PE changes into KE. Lesson 3 asks a new question: when energy changes form, does the total change? Learners find that in an ideal system the total stays constant. This gives the first big idea for the driving question: when Mama Kavata or Mwende lifts the bucket, the energy is not made from nothing. Lesson 4 will deal with real systems and energy losses, and Lessons 5 to 8 will connect work, force and the pulley system.
Safety Notes	The falling bucket is a paper-cup or small sealed-bag demonstration with a small mass, dropped only over a clear floor area; learners stand back 2 m. The pendulum demo uses a small bob (a stone or nut) on a string, hung from a secure stand; keep faces away from the swing path. Keep walkways clear and bags off the floor during computer work. Learners share devices and handle them gently.

B. LESSON OVERVIEW

Learners start by predicting what happens to the total energy of a swinging pendulum, and of the bucket released from the top of the Kitui well, as PE turns into KE. In the Observe phase, groups use PhET Pendulum Lab and Mr Otieno's string-and-stone pendulum demonstration to collect data, then complete the free-fall table for the 20 kg bucket released from 8 m and add KE + PE at each height. In the Explain phase, learners use their data to state the conservation of energy principle for ideal systems and link it to the pulley system: the energy a person puts in has to go somewhere. They add new questions to the DQB and revise their model with a total-energy bar for the bucket (1600 J at every height) and a note that the energy Mama Kavata and Mwende use comes from the chemical energy in food such as ugali and githeri.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Learners sit in groups of 4 and look at a projected or sketched	Say: 'Last lesson we calculated KE and PE and watched PE change	Predict-Observe-Explain (POE) with a Prediction Wall: committing	Check the written predictions and bar sketches. Indicators: (1)

Conservation of energy Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) Search ARES for similar readings	image of a pendulum bob (0.5 kg) pulled aside 0.2 m high, and of the 20 kg bucket held at the top of the Kitui well (8 m above the water). Each learner silently writes a prediction in their notebook for both: 'As it moves down, PE goes down and KE goes up. What happens to the TOTAL (KE + PE)? Does it grow, shrink or stay the same? Why?' They draw a quick bar sketch showing the PE and KE bars at the top, middle and bottom. They share in their groups and record one group prediction on a sticky note, and then place it on the Prediction Wall.	into KE. Today's question is different: when the form changes, what happens to the total?' Show the pendulum 1 J example from Lesson 2 (PE at the top 1 J, KE at the bottom 1 J) without telling the answer. Say: 'Predict for the pendulum and for the bucket in Kitui. Write your own idea first. No talking for 2 minutes.' WAIT TIME 10-15 seconds after asking: 'What do you think happens to the total?' Cold-call 4 learners (from different groups) to share their prediction WITHOUT saying whether it is right. Ask: 'Who disagrees? What reason would you give?' Record 2 or 3 competing ideas on the board (e.g. 'total grows', 'total shrinks', 'total stays the same'). Circulate and note which groups hold the idea that energy is 'used up' as the bucket falls.	to a written prediction first creates a need to know and makes learners' existing ideas (energy is 'used up' or 'created' by motion) visible, so the evidence in the Observe phase can confirm or challenge them.	learner states a clear prediction about the total, not only about KE or PE separately; (2) learner gives a reason; (3) bar sketches show KE and PE changing in opposite directions. Tally on a class chart how many predict 'total grows', 'shrinks' or 'stays the same'. Note learners who say energy is 'lost' or 'used up' as an idea to revisit in Lesson 4.
Observe Phase  VIDEO: Conservation of energy Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Conservation of Energy (Flexbook) Source: CK-12 Search ARES for similar readings	Part A (PhET Pendulum Lab, about 20 minutes): In pairs on a shared device, learners open the Energy graph (bar chart) view. They set the mass to 0.5 kg and the starting height equivalent to 0.2 m, release the bob, and watch the PE, KE and total energy bars with friction set to NONE. They pause at 4 positions (top, halfway down, bottom, halfway up) and record PE, KE and total in a table. Total should read about 1 J every time. They then change the mass (e.g. 1.0 kg) and the starting height and repeat once, noting the total changes only when the starting height or mass changes, not during the swing. Then they turn friction ON and describe what happens to the total bar over time (the swing slowly dies down; note this for Lesson 4). Part B (teacher demonstration and calculation,	Before Part A say: 'Your job is to collect evidence about the total energy. Watch the TOTAL bar, not only PE and KE.' Distribute roles: operator, recorder, checker, reporter (rotate them). Circulate and ask each group: 'What is the total at the top? At the bottom? What do you notice?' WAIT TIME 10-15 seconds after each question. If a group sees 1.0 J but reads 0.98 J because of rounding, say: 'Why might the numbers differ slightly? Is that a real change or a reading issue?' For Part B, say: 'Mr Otieno will not tell you the answer. Calculate it.' Model the first row (8 m) on the board: $PE = 20 \times 10 \times 8 = 1600 \text{ J}$. Cold-call 3 learners to give the PE at 4 m and the KE at 4 m, then ask: 'What is the total at 4 m?' Support learners who use $g = 9.8$ by reminding: 'In this class, $g = 10 \text{ N/kg}$.' Keep the speed column	Data table and graph analysis (Patterns in data): learners gather quantitative evidence from two different systems (the pendulum and the bucket) and look for what stays the same while other quantities change. The flat total line against the rising and falling PE and KE lines makes the pattern visible and builds the claim from evidence, using the practice of analysing and interpreting data.	Check tables and graphs. Indicators: (1) PE values at 8, 6, 4, 2 and 0 m are 1600, 1200, 800, 400 and 0 J; (2) KE + PE is shown as 1600 J in every row; (3) the total line on the graph is flat and the PE and KE lines mirror each other; (4) the pendulum table shows a constant total of 1 J. Ask the reporter of 2 groups: 'What stays the same? What changes?' Look for correct units (J) and correct use of $g = 10 \text{ N/kg}$.

	about 25 minutes): Mr Otieno shows a string-and-stone pendulum and marks the highest point of each swing with chalk to show the bob returning to nearly the same height. Learners then complete the free-fall table for the 20 kg bucket released from the top of the Kitui well (8 m above the water surface, no losses): at 8 m PE = 1600 J, KE = 0 J; at 6 m PE = 1200 J, KE = 400 J, $v = 6.3$ m/s; at 4 m PE = 800 J, KE = 800 J, $v = 8.9$ m/s; at 2 m PE = 400 J, KE = 1200 J, $v = 11.0$ m/s; at 0 m PE = 0 J, KE = 1600 J, $v = 12.6$ m/s. Learners calculate PE = mgh using $g = 10$ N/kg, check KE = $\frac{1}{2} \times 20 \times v$ squared from the given speeds, and add a column for KE + PE. They graph PE, KE and total against height on one set of axes (3 lines).	given ($v = \sqrt{2gh}$) is not derived today), so that the focus stays on totals. Ask: 'What would the graph of the total look like? Sketch it before you plot.' Do not state the principle yet.		
Explain Phase  VIDEO: Conservation of energy - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Conservation of Energy Practice Source: CK-12  Search ARES for similar readings	Learners meet in their groups with their tables and graphs and write a CER paragraph (Claim, Evidence, Reasoning) answering: 'In an ideal system with no friction, what happens to the total energy (KE + PE) as an object falls or swings?' Claim: the total stays constant. Evidence: they cite specific numbers (1600 J at every height for the bucket; 1 J for the pendulum). Reasoning: energy changes form from PE to KE (and back for the pendulum) but is not created or destroyed. Then they take part in a whole-class discussion, comparing their Predict-phase ideas with the data, and revising any incorrect ideas. Finally they apply the idea to the Kitui phenomenon in a short talk: 'Mama Kavata or Mwendu lifts the bucket 8 m. The bucket gains 1600 J of PE. If energy cannot be	Say: 'Look at your data, then complete: In an ideal system, the total energy ... because ...' WAIT TIME 10-15 seconds before calling on anyone. Cold-call 3 groups to read their claim and evidence; ask the others: 'Do you agree? Which number supports it?' Introduce the term for the pattern: 'This is the principle of conservation of energy: energy is never created or destroyed, only transformed or transferred.' Revisit the Prediction Wall: 'Whose prediction was supported? Whose idea changed?' Probe misconceptions: 'Some of you said energy is used up when the bucket falls. Where does the PE go?' (it becomes KE). Then link to Kitui: 'Mwendu pulls the rope and the bucket gains 1600 J. Where does that energy come from?' Cold-call 2 learners. If someone says 'the pulley', ask:	Claim-Evidence-Reasoning (CER) with Argument from Evidence: learners must connect numerical evidence to a general principle, and defend it against peers. Applying the principle to the Kitui bucket (a transfer task) builds understanding that the principle explains the phenomenon, rather than being just a result from two lab systems.	Collect CER paragraphs. Indicators: (1) the claim states that total energy stays constant in an ideal system; (2) evidence uses at least two specific values (e.g. 1600 J at 8 m and at 0 m); (3) reasoning mentions transformation of PE into KE rather than energy being used up or created; (4) learner identifies that the energy for lifting the bucket comes from the person's body (chemical energy from food), not from the pulley. Use a 3-level check: Not yet (claim only), Developing (claim and number), Secure (claim, number and reasoning).

	created, where did that 1600 J come from?' Learners suggest answers (the chemical energy in her body from ugali and githeri), and they then discuss whether the pulley could 'make' extra energy (no). They note an open question: if the pulley makes it easier, is something else different, and is any energy wasted? (to be developed in Lessons 4 to 8).	'Can a rope and a wheel make energy? What does your data say about creating energy?' Say clearly that the word 'ideal' means no friction, and that real systems are the focus of Lesson 4. Do not calculate the pulley effort or work today (Lessons 5, 6 and 8).		
Driving Question Board (DQB) Creation  VIDEO: Work and energy (part 2) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Energy Skate Park: Basics Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners return to the Driving Question Board started in Lesson 1: 'How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process?' In groups, they write on sticky notes (a) one thing they now know from today's evidence (e.g. 'Total energy stays 1600 J as the bucket falls in an ideal system') and (b) one new question (e.g. 'Why did the pendulum slow down when friction was on?', 'If the energy is conserved, why is the pulley easier?', 'Where does the energy go in a real fall?'). They place notes on the DQB under headings such as 'Energy conservation', 'Energy loss' and 'Pulleys and effort', and mark which questions are answered and which will be investigated in later lessons.	Say: 'Choose one question that today's evidence helped you answer, and one that it made you wonder about.' Give 3 minutes of silent writing, then WAIT TIME 10-15 seconds before asking for volunteers. Cold-call 4 learners to place and read a note and to explain which category it fits. Group similar questions (e.g. friction/energy loss questions that connect to Lesson 4; effort versus distance questions that connect to Lessons 5 to 8). Say: 'Some of your questions we will answer in the next lessons. Keep watching the board.' Mark the DQB with a tick next to answered questions, e.g. 'Where does the energy come from?' (food, chemical energy), but leave 'Why is it easier?' and 'Is energy lost?' open.	Driving Question Board (question tracking): recording what has been learned and what remains unanswered keeps learners' own questions steering the unit, and it shows how today's evidence (conservation) moves the class closer to answering the driving question while revealing the next gaps (losses and effort).	Review the notes on the board. Indicators: (1) at least one new question per group that refers to energy conservation, losses or effort; (2) questions use terms such as KE, PE, total energy and friction correctly; (3) learners sort their questions into categories and can say which lesson might answer them. Note any group that has only recall questions, and prompt them with: 'What do you still not understand about why the pulley feels easier?'
Model Building Phase  VIDEO: Introduction to energy <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES	Learners take out the initial model they began in Lesson 1 (a diagram of the Kitui well showing the bucket, the woman or child, the rope and pulley, and energy ideas). Individually, then in pairs, they revise it to include: (1) a bar chart or labelled arrows showing PE and KE of the bucket at 8 m, 4 m and 0 m, with a 'Total = 1600 J' label at every height for the ideal fall; (2) an arrow showing the	Say: 'Use a new colour to show what you now understand about energy. Your model must show where the energy goes and where it comes from.' Circulate and ask: 'Where is the 1600 J at the top? At the bottom? How does your model show that the total is the same?' WAIT TIME 10-15 seconds. Press learners who show the pulley as a source of energy: 'What does the conservation principle say about	Cumulative model revision (Developing and using models): learners incorporate new evidence into a visual model and mark what changed, so the model becomes a record of their growing explanation. Leaving the pulley effort as an explicit open gap shows models as tools that are revised as new evidence arrives.	Check the revised models. Indicators: (1) the model shows PE and KE for the bucket with a constant 1600 J total; (2) chemical energy from food is shown as the source of the energy used to lift; (3) the conservation statement is written correctly, using 'transformed or transferred' language; (4) changes are marked in a different colour; (5) an open question on the pulley effort or

for similar videos  HTML: Conservation of Energy (Flexbook) Source: CK-12  Search ARES for similar readings	chemical energy from food (ugali, githeri) in Mama Kavata's or Mwende's body transferring energy to the bucket as PE when lifted; (3) a statement box: 'In an ideal system, total energy is conserved: energy is transformed or transferred, never created or destroyed'; (4) a question mark beside the pulley with 'Why is the effort smaller? Is any energy wasted?' as an open item to revise later. Learners use a different colour pen to show what changed from their earlier model. A few pairs present to the class.	that?' Cold-call 3 pairs to present one change they made and the evidence that led to it. Close with: 'Our model now explains why the total energy stays fixed. What it does not yet explain is why the pulley makes the effort smaller, or whether any energy is wasted in a real system. We will tackle losses next lesson.' Ask learners to keep the revised model in their folders.	losses is recorded. Learners who show energy 'used up' or created by the pulley need a one-to-one conversation using their own data table.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners predict that the total would stay constant before the evidence, or did they hold the idea that energy is used up as the bucket falls? Which data convinced them to change their minds? (2) Could learners correctly complete the KE + PE column with $g = 10 \text{ N/kg}$, or did many use 9.8 or forget to use mass? (3) Did the PhET friction-on observation raise questions about energy loss that will help launch Lesson 4? (4) In the Kitui transfer talk, how many learners identified the body's chemical energy from food as the source, rather than the pulley? (5) Which groups or learners were not heard during cold-calling, and how will you include them next lesson? (6) If there was limited access to devices, did the string-and-stone pendulum and the table give enough evidence? Note adjustments for Lesson 4, which deals with real systems, energy losses and the measured fall speed of 11 m/s.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the PhET Pendulum Lab and in the free-fall table for the 20 kg bucket released from 8 m, we observed that PE decreases and KE increases as the object moves down, while the total KE + PE stays the same: 1600 J at every height for the bucket and 1 J for the 0.5 kg pendulum bob raised 0.2 m. With friction turned on in the simulation, the swing slowly died down.
What did I learn?	We learned that in an ideal system (no friction) the total energy stays constant. Energy is never created or destroyed, only transformed (PE to KE and back) or transferred. We also learned that the energy to lift the bucket in Kitui comes from the chemical energy in the person's body from food such as ugali and githeri, not from the pulley.
How does this explain the phenomenon?	The bucket gains 1600 J of PE when it is lifted 8 m, so this energy must come from somewhere, because energy cannot be made from nothing. The pulley cannot create energy. We still need to explain why the pulley makes the effort smaller and whether any energy is lost in a real system; we will investigate these questions in the next lessons.

LESSON 4 (40 minutes (double lesson 80 minutes if practical sliding demo is included)): Conservation of Energy in Applications: Where Does the Bucket's Energy Go in a Real Fall?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply conservation of energy to solve problems on falling and sliding objects, and compare an ideal fall with a real fall of the 20 kg bucket to account for energy transferred as heat and sound.
Knowledge	By the end of the lesson, the learner should be able to: (a) use $KE + PE = \text{constant}$ to find the speed of a falling object in an ideal system, (b) compare the ideal fall of the bucket ($KE = 1600 \text{ J}$, $v = 12.6 \text{ m/s}$) with the real fall ($v = 11 \text{ m/s}$, $KE = 1210 \text{ J}$), (c) account for the 390 J difference as energy transferred to heat, sound and air/rope friction, (d) explain that 'lost' energy is transferred to the surroundings, never destroyed.
Skills	Developing and using models (energy bar charts); Analyzing and interpreting data (ideal versus real table); Using mathematics and computational thinking ($KE = \frac{1}{2}mv^2$, $PE = mgh$, $v = \sqrt{2gh}$); Constructing explanations; Communication and collaboration in groups.
Attitudes	Curiosity about where energy goes; Responsibility in sharing roles within the group; Respect for others' ideas during discussion; Appreciation of energy-saving in everyday machines.
Key Inquiry Question	1. How do we use energy in our day-to-day life? 2. How do machines make work easier?
Purpose in Storyline	Lesson 3 showed that in an ideal system $KE + PE$ stays constant (1600 J at every height of the free-falling bucket). Real systems like the well in Kitui do not behave ideally. This lesson gives learners the evidence that some energy is transferred to the surroundings as heat and sound. This prepares them in later lessons to explain why the pulley system wastes 400 J while the bucket still gains 1600 J, and it advances the driving question about where the energy goes.
Safety Notes	If doing the sliding demo: use a smooth plank or tray on the floor, keep feet clear of the sliding object, use light objects (small block or pencil case), do not drop heavy objects. Keep the walkway clear. No real buckets are dropped; the fall is analysed from data and video.

B. LESSON OVERVIEW

Mr Otieno's learners return to Mama Kavata's well in Kitui. Lesson 3 showed that for a bucket falling freely with no losses, $KE + PE = 1600 \text{ J}$ at every height and it would hit the water at 12.6 m/s. In this lesson learners first predict what would happen in a real fall, then analyse data showing that the measured speed is only 11 m/s ($KE = 1210 \text{ J}$). Using conservation of energy they find that $1600 - 1210 = 390 \text{ J}$ went to heat, sound and air/rope friction. They extend the idea to a sliding object on rough and smooth surfaces, then add this evidence to the DQB and revise the class model. Key message: energy is not destroyed, it is transferred to the surroundings. The pulley system and its efficiency are left for later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the free-fall table	Display the free-fall table on the	Predict-Observe-Explain with	Look for: learners who draw the

 VIDEO: Conservation of energy Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Conservation of Energy (Flexbook) Source: CK-12  Search ARES for similar readings	from Lesson 3 (bucket released from 8 m, PE 1600 J at the top, KE 1600 J and $v = 12.6$ m/s at the water, no losses). Individually they write a prediction in their notebooks: 'The bucket is dropped down the real well in Kitui. Will it hit the water at 12.6 m/s, faster, or slower? Where would the missing energy go?' They sketch a quick energy bar chart for the top and the bottom of the fall. They then share with a partner.	board. Say: 'Yesterday we found that in an ideal fall the bucket reaches the water at 12.6 m/s. Mama Kavata's well is not ideal. Think quietly: will a real bucket reach the same speed?' Give 2 minutes of silent writing. WAIT TIME of 10-15 seconds after asking: 'Where might some of the energy go if the bucket is slower?' Cold-call 4 learners (2 who predict 'same', 2 who predict 'slower') and ask: 'What makes you think so?' Record predictions in two columns on the board without judging. Do not give the answer.	Think-Pair-Share: committing to a prediction first exposes the common idea that energy 'disappears' or that real falls match the formula exactly, so the later data creates a productive conflict.	bottom bar as full KE (ideal thinking) versus those who add a bar for 'other energy'. Tally predictions on the board. Listen for words such as 'friction', 'air', 'heat', 'sound'. Note learners who think energy 'disappears' for follow-up.
Observe Phase  VIDEO: Work/energy problem with friction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Conservation of Energy Practice Source: CK-12  Search ARES for similar readings	Part A (15 min): In groups of 4 learners study the data card: the real fall of the 20 kg bucket from 8 m measured 11 m/s at the water (a slow-motion video clip of a bucket falling or the data card if no video is available). They calculate: (1) PE at the top = $20 \times 10 \times 8 = 1600$ J; (2) real KE at the bottom = $\frac{1}{2} \times 20 \times 11 \times 11 = 1210$ J; (3) ideal KE = 1600 J, $v = 12.6$ m/s; (4) difference = $1600 - 1210 = 390$ J. They fill in an Ideal versus Real table. Part B (10 min): Sliding demo or data: a block or pencil case is released from the top of a smooth tilted plank or tray and slides to the bottom, then pushed across a rough floor (dirt/concrete) until it stops. Learners touch the plank/floor and the block after repeated slides and note warmth and any sound. They discuss: 'Where did the motion energy of the block go when it stopped?'	Circulate with the questions: 'What is the total energy at the top?' 'Is the KE at the bottom equal to 1600 J? How do you know?' Remind: 'Use $g = 10$ N/kg and show every step: substitution, answer with unit.' For the sliding demo say: 'Rub your hands together for 10 seconds. What do you feel? Now compare with the block.' WAIT TIME of 10-15 seconds after: 'The block has stopped. It had KE a moment ago. Where is that energy now?' Cold-call 3 groups to report their 390 J result before moving on. Support weaker groups by giving the formula $KE = \frac{1}{2} mv^2$ squared on the board. Challenge fast groups: 'Calculate the speed the bucket would have if all 1600 J became KE, and show it equals 12.6 m/s.'	Data table comparison (ideal versus real): placing the two columns side by side makes the energy gap visible and measurable. Linking the sliding observation (warmth, sound) to the numeric gap helps learners connect abstract joules to physical evidence.	Check that groups: (a) correctly calculate 1210 J and 390 J, (b) use the units J and m/s, (c) identify heat and sound as destinations of the 390 J. Observable indicator: group tables show KE + other energy = 1600 J. Tick a checklist for groups that link the warm block or sound to energy transfer.
Explain Phase  VIDEO: Thermal energy from friction Source: Khan	Learners work in the same groups to build an energy bar chart for the real fall at three points: top (PE 1600 J), halfway at 4 m (ideal: PE 800 J, KE 800 J; real: some	Say: 'Total energy is conserved, but not all of it stays as motion energy. Who can tell me, in one sentence, why the real bucket is slower?' Give WAIT TIME of 10-15	Claim-Evidence-Reasoning with energy bar charts: structured argument forces learners to justify with numbers (1600, 1210, 390 J), and bar charts model the idea of	Look for explanations that state the total remained 1600 J, naming heat and sound as destinations. Red flags: 'the energy was destroyed' or 'friction used up the

Academy (English - US curriculum) Search ARES for similar videos HTML: Conservation of Energy (Flexbook) Source: CK-12 Search ARES for similar readings	energy already shared to the surroundings) and the water (KE 1210 J plus 390 J to heat, sound and air/rope friction). They write a Claim-Evidence-Reasoning statement: 'Claim: the energy of the bucket was conserved. Evidence: ... Reasoning: ...' They solve one extra problem: a 2 kg block slides from rest down a smooth slope 0.5 m high with no friction; find its speed at the bottom ($PE = 2 \times 10 \times 0.5 = 10 \text{ J}$, so $KE = 10 \text{ J}$, $10 = \frac{1}{2} \times 2 \times v^2$, $v = 3.2 \text{ m/s}$). Then a real case: it reaches the bottom at 3.0 m/s; $KE = \frac{1}{2} \times 2 \times 9 = 9 \text{ J}$, so 1 J was transferred as heat and sound. They finish by linking back: 'What does this tell us about the pulley system in Mama Kavata's well?'	seconds, then cold-call 3 learners and build on responses by asking 'Who can add evidence from our table?' Correct the misconception that energy is destroyed: 'Energy is never destroyed. When we say lost, we mean transferred to the surroundings as heat and sound, where we cannot easily use it.' Ask: 'Mwende pulls the rope over the pulley and it warms slightly. What might that tell us?' Do not give the pulley system numbers; say: 'We will measure exactly how much in a later lesson.' Select two groups to present bar charts on the board.	energy being shared among stores while the total stays 1600 J.	energy'. Check calculations of 3.2 m/s and 1 J. Use exit-ticket style mini whiteboards: 'Write the energy equation for the real fall' ($1600 \text{ J} = 1210 \text{ J} + 390 \text{ J}$). Count how many learners show it correctly out of the class.
Driving Question Board (DQB) Creation VIDEO: Conservation of energy Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Conservation of Energy Practice Source: CK-12 Search ARES for similar readings	Learners revisit the class DQB started in Lesson 1. Each group writes on sticky notes: (1) one thing learned today that helps answer 'How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process?' (for example: 'Some energy goes to heat and sound, so real machines lose some energy'), and (2) one new question (for example: 'How much energy is lost in the pulley system?', 'Does a pulley make friction heat too?'). They place notes next to the matching questions and mark any questions now partly answered.	Say: 'Look at the board. Which of our questions did today's evidence help with?' Allow WAIT TIME of 10-15 seconds before taking answers. Cold-call 4 groups to read one sticky note each and explain where to place it. Ask: 'Is this question answered fully or partly? What evidence do we still need?' Highlight any question about energy lost in the pulley and say: 'We will come back to this when we study work.' Keep the questions on the board for Lesson 5.	Driving Question Board tracking: publicly linking new evidence to the original questions shows learners how knowledge builds over lessons and makes their curiosity visible and drives the next investigation.	Check that notes link evidence to questions (not just copied facts). Observable indicators: new questions refer to the pulley, friction or heat; at least one question per group. Record which questions remain open for planning Lesson 5.
Model Building Phase VIDEO: Work/energy problem with friction	Learners revise their individual models of the bucket in Mama Kavata's well from Lesson 1. They add: (1) an energy bar chart for the falling bucket showing PE 1600 J changing into KE 1210 J plus 390 J to heat, sound and air/rope	Say: 'Use a different colour to show what has changed since Lesson 3. A model must explain what we have seen.' Circulate and ask: 'Where is the 390 J in your model?' 'How do you show the total stays the same?' WAIT TIME	Cumulative model revision with colour coding: updating the same model each lesson makes changes in thinking visible and links all evidence (ideal fall, real fall, sliding) to the anchoring	Rubric check: model includes (1) PE 1600 J at the top, (2) KE 1210 J at the bottom, (3) 390 J arrow to heat/sound, (4) statement of conservation. Indicators of mastery: all four present. Partial: missing the 390 J destination.

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Energy Skate Park: Basics <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	friction; (2) a labelled arrow showing energy going to the surroundings; (3) a note that 'total energy is always conserved'. They add a small sketch of Mwende pulling the rope with a note saying 'some energy may be lost as heat in the pulley and rope (to be measured later)' using a different coloured pen to show what is new. They write a 2-sentence summary at the bottom: 'In real systems, energy is conserved but some is transferred as heat and sound. This is why ...'	of 10-15 seconds before answering learners' questions to let them reflect. Cold-call 3 learners to explain one revision. Collect models for review. Close with: 'Energy comes from food like ugali and githeri in our bodies; where do you think the energy of Mwende's pull ends up?'	phenomenon.	Note learners needing support before Lesson 5.
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D. TEACHER REFLECTION

Which learners still believe energy is destroyed or 'used up' rather than transferred? Did the sliding block observation (warmth, sound) help them connect numbers to physical evidence? Did groups use $g = 10 \text{ N/kg}$ and show full working for KE and PE? Which DQB questions remain open, particularly about the pulley's energy loss, and how will Lesson 5 (work $W = Fd$) address them? What adjustments are needed for learners who struggled with $v = \sqrt{2gh}$ or with the square in KE?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the real bucket fall measured 11 m/s at the water, not the ideal 12.6 m/s. Its KE was 1210 J instead of 1600 J. A sliding block became warm and made sound and stopped on a rough surface.
What did I learn?	We learned that in real systems the total energy is still conserved, but some energy is transferred to the surroundings as heat and sound through friction. 'Lost' energy is not destroyed. The 1600 J of PE became 1210 J of KE plus 390 J of heat, sound and air/rope friction.
How does this explain the phenomenon?	We can explain that the real bucket is slower than the ideal one because 390 J of its energy was transferred to heat and sound, so energy is conserved overall. This suggests that in Mama Kavata's pulley system some energy will also be transferred as heat by friction, which we will measure later.

LESSON 5 (40 minutes (1 double period or 1 lesson plus 5-minute extension)): Work: $W = Fd$ – What Does Mama Kavata's Lift Really Do?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners define work as force times distance moved in the direction of the force, calculate the work done in lifting the 20 kg bucket 8 m from the well, and tell positive work from negative work using lifting and lowering examples.
Knowledge	Work is done when a force moves an object in the direction of the force: $W = Fd$ (joules). Direct lift: $200\text{ N} \times 8\text{ m} = +1600\text{ J}$ by Mama Kavata. Gravity does -1600 J during the lift (force opposite to motion), so net work = 0 J at steady speed. Lowering the bucket 8 m slowly: work by the woman = -1600 J , work by gravity = $+1600\text{ J}$. If there is no movement, or the force is at right angles to the motion, no work is done in the physics sense.
Skills	Predicting, calculating with $W = Fd$ using $g = 10\text{ N/kg}$, interpreting signs of work, constructing explanations from evidence (NGSS SEP: Using Mathematics and Computational Thinking; Analyzing and Interpreting Data; Constructing Explanations), collaborating in groups.
Attitudes	Curiosity about why 'effort' in daily language differs from 'work' in physics; appreciation of the effort of rural water-drawing; respect for others' ideas; responsibility in group roles.
Key Inquiry Question	1. How do we use energy in our day-to-day life? 2. How do machines make work easier?
Purpose in Storyline	Lessons 1-4 built energy ideas (KE, PE, conservation, losses) and the joule. Lesson 5 introduces work as the way energy is transferred by a force. It gives the quantity (1600 J for the direct lift) that Lesson 6 will compare with the pulley system. It does not yet analyse the pulley or power; learners add 'work = Fd ' and 'signs of work' to the DQB and model.
Safety Notes	Use a light school bag or 2 kg load for the lifting activity, lifted only 1 m from a bench to chest height. Keep feet clear of the load. Do not hang loads over heads. Wipe any spilt water on the floor immediately. Lift using legs, with a straight back.

B. LESSON OVERVIEW

Mr Otieno opens in Kisumu by recalling Mama Kavata in Kitui lifting the 20 kg bucket (weight 200 N) up the 8 m deep well. Learners predict how to measure 'how much work' she does. They then gather evidence from a hands-on lift (a 2 kg load raised 1 m), a data table and short worked examples to build $W = Fd$. The key idea: work is a force moving something in the direction of the force. Learners calculate the direct lift: $200\text{ N} \times 8\text{ m} = 1600\text{ J}$, matching the PE gain from the Lesson 1-2 table. They examine work signs: the woman does $+1600\text{ J}$, gravity does -1600 J , net work 0 J at steady speed, and lowering reverses the signs. They add the new questions to the DQB and revise their model with a work arrow. The pulley comparison is left to Lesson 6.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Work and the	Learners look at the photo or sketch of Mama Kavata lifting the bucket 8 m by hand. Individually	Say: 'Remember Mama Kavata in Kitui and her 20 kg bucket. In Lessons 1 to 4 we learned that	Eliciting prior ideas using Predict-Write-Pair-Share: writing individually first makes each	Observable indicators: (1) learner predictions in notebooks showing everyday meaning ('work =

work-energy principle Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Applications: Work and Force (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	they write in their notebooks: (a) 'What do you think we mean by the word work in physics?' and (b) 'Mama Kavata holds the bucket still for 10 seconds with her arms shaking. Does she do work on the bucket? Why or why not?' They also predict a number: 'How many joules do you think she transfers to the bucket when she lifts it 8 m?' They then share with a partner and mark each prediction with a confidence level (1-3).	energy can be stored or transformed and measured in joules. Today we ask: how is energy actually moved into the bucket?' Display the picture. Say: 'Write your own idea first. There are no wrong predictions today.' WAIT TIME: 15 seconds of silent thinking before writing starts. Walk around and note on a clipboard three different ideas (everyday 'work' = tiredness; force only; force and distance). Cold-call 4 learners: 'Tell us your prediction, and what made you decide?' Ask a follow-up: 'Who disagrees, and why?' WAIT TIME: 10 seconds after each follow-up. Record the predictions on the board without judging. Say: 'We will test these ideas using evidence.'	learner commit to an idea, so later evidence can be compared with it and shift thinking. Naming the everyday meaning of 'work' (tiredness, jobs) exposes the difference from the physics meaning, which sets up conceptual change.	effort/tiredness') versus physics-related ideas (force, movement); (2) at least 4 learners can give a reason for their prediction for the holding-still question; (3) the teacher tallies on the board how many predict 'she does work when holding still' (likely a majority). Look for the misconception that holding a load still is work in the physics sense.
Observe Phase  VIDEO: Work as the transfer of energy Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: 17 Forces At Work <i>Related topic: not an exact match for this lesson.</i> Source: iicba Search ARES for similar readings	In groups of four (roles: lifter, measurer, recorder, reporter), learners do a 'Mini-Well' lifting investigation. Using a 2 kg load (a bag of maize flour or sand, weight 20 N) they lift it slowly and steadily 1 m from the floor to a bench. They record: force needed (about 20 N, read from the load label or a spring balance if available), distance (1 m, measured with a metre rule), and calculate force x distance. Next they hold the load still for 5 seconds and record the distance moved (0 m). Then they carry the load 2 m horizontally at a steady height and note that the lifting force is vertical while the motion is horizontal. Finally, they study a data table linking the real bucket: Force 200 N, distance 8 m, and the PE table (0 m = 0 J up to 8 m = 1600 J). They compute $F \times d$ for 2 m (400 J), 4 m (800 J), 6 m (1200 J) and 8 m (1600 J) and compare the answers with the PE	Distribute materials and say: 'This is our mini-well. Mama Kavata lifts 200 N over 8 m; you will lift 20 N over 1 m. The rule you discover should work for both.' Safety reminders: 'Light loads only, bend your knees, keep feet clear.' Circulate and ask each group: 'What is the force doing? Which way is it pointing? Which way is the load moving?' WAIT TIME: 10 seconds after each question before prompting. Ask the measurer: 'What was the distance moved while you held it still?' Ask: 'Did the load gain height when you carried it sideways?' After 12 minutes, call groups back. Cold-call 3 groups to report their force x distance result for the lift (expected 20 J). Then project the bucket table and ask: 'What do you notice when you multiply 200 N by each height?' WAIT TIME: 15 seconds. Cold-call 4 learners to state a pattern. Confirm that 200 N x 8 m	Data patterns through hands-on measurement followed by Noticing and Wondering on a data table: learners construct the rule $W = Fd$ from their own measurements and from the bucket values, rather than being given it. Comparing the holding-still and sideways-carry trials with the lift isolates the role of 'distance in the direction of the force'.	Observable indicators: (1) groups correctly record $F = 20\text{ N}$, $d = 1\text{ m}$ and compute 20 J; (2) the recorder writes 0 J (or 'no movement, no work') for holding still; (3) at least 3 of 4 groups correctly calculate 400, 800, 1200 and 1600 J in the bucket table; (4) learners point to the arrow of the force and the arrow of motion on their sketches. Teacher checklist: who correctly says that carrying sideways involves no work by the lifting force?

	column.	= 1600 J matches the PE gain. Do not yet introduce the pulley.		
Explain Phase  VIDEO: Work and the work-energy principle Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Forces and Motion: Basics <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners build a claim-evidence-reasoning (CER) explanation. Claim: 'Work is done when ...'. Evidence: their measurements (20 N x 1 m = 20 J; holding still = 0 J; bucket 200 N x 8 m = 1600 J). Reasoning: why direction matters. Then, in pairs, they analyse work signs with a sign table: (1) Mama Kavata lifting the bucket up 8 m at steady speed: her force is up and motion is up, so work = +1600 J. (2) Gravity (200 N down) while the bucket goes up: force opposite to motion, so work = -1600 J. (3) Net work = +1600 + (-1600) = 0 J, which is why the bucket's speed does not change (KE constant) while its PE rises by 1600 J. (4) Lowering the bucket slowly 8 m by rope: the woman's force is up while motion is down, so work = -1600 J; gravity's work = +1600 J. Learners then solve two quick problems: Mwende (30 kg) climbs onto a stool 0.5 m high, so the work done against gravity is 300 N x 0.5 m = 150 J; and a farmer in the Rift Valley pushes a wheelbarrow with a 60 N force through 10 m: work = 600 J. They finish with a short note stating the definition, formula and units (1 J = 1 N x 1 m).	Say: 'Use your evidence. Complete the sentence: work is done when ...' WAIT TIME: 15 seconds before pairs begin writing. Circulate and press: 'How does your evidence support that? What happened when the load did not move?' Draw the bucket on the board with arrows: force up (green), gravity down (red), motion up. Ask: 'Which arrows point the same way? Which point opposite?' WAIT TIME: 10 seconds. Cold-call 3 learners to state the sign of the woman's work and gravity's work. Ask: 'If the net work is zero, why did the bucket still gain 1600 J of PE?' Guide: the woman transfers energy into the bucket (+1600 J) while gravity's negative work shows the energy being stored as PE. Use the phrase: 'Positive work gives energy to an object; negative work takes energy from it or stores it as the force opposes the motion.' Take care to keep consistent with the established facts: the net work is 0 J so KE is unchanged. Then ask: 'What changes for the lowering case?' Cold-call 3 pairs to state the new signs (-1600 J woman, +1600 J gravity). Clarify misconceptions: 'Negative does not mean wasted or smaller; it means force and motion are opposite.' Note that the effect of the pulley on the distance and force will be examined in Lesson 6. Tell learners: 'Next lesson we compare this 1600 J with Mwende's pull on the pulley system.'	Claim-Evidence-Reasoning (CER) with a sign-comparison table: learners justify the definition of work with their own evidence, then use force and motion arrows to reason about positive and negative work. The multiple-representation approach (arrows, table, numbers) makes the abstract sign convention concrete and links it to the energy ideas of Lessons 1-4 without re-teaching them.	Observable indicators: (1) CER statements mention both force and movement in the direction of the force; (2) learners correctly give +1600 J, -1600 J and 0 J for lift, gravity and net; (3) learners reverse the signs for lowering; (4) answers to the stool (150 J) and wheelbarrow (600 J) problems with units. Exit check: hands up for 'work is done when a person holds a load still' (should now be false). Misconception watch: 'negative work means no work' or 'net 0 J means nothing happened'.
Driving Question	Learners revisit the class DQB started in Lesson 1. In groups,	Say: 'Look back at our DQB. Which questions did today's	Driving Question Board as a public record of growing understanding:	Observable indicators: each group posts at least one evidence-based

Board (DQB) Creation  VIDEO: Work as the transfer of energy Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Applications: Work and Force (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	they write on sticky notes (a) one thing they now know about the driving question 'How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process?' (for example: 'Lifting the bucket by hand needs 1600 J of work, 200 N over 8 m'), and (b) one new question (for example: 'If the work to lift is 1600 J, how can Mwende pull with less force?', 'Does the pulley change the work?', 'Where does the extra energy go if the rope is longer?'). They place notes next to existing questions and mark which earlier questions are now partly answered.	evidence help us answer?' WAIT TIME: 10 seconds. Ask each group to bring one 'now we know' note and one new question. Cold-call 4 groups to read and place their notes. Ask: 'Which question will we test next?' Highlight any question about force and distance with a star and say: 'Lesson 6 will compare hand lifting and the pulley system using the work we have calculated today.' Do not answer questions about efficiency, power or mechanical advantage; say: 'Park that one, we will get to it with more evidence.' Mark with a tick the DQB question 'What is work?' as partly answered.	moving from questions to partial answers makes learning visible, shows how today's evidence (work = Fd) links to the unresolved pulley puzzle, and motivates the next investigation.	'now we know' note using F, d and joules; at least two new questions concern comparing the pulley with the direct lift; learners can explain why a question is only partly answered. Teacher records which groups still think 'effort' and 'work' mean the same thing.
Model Building Phase  VIDEO: Power <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: 17 Forces At Work <i>Related topic: not an exact match for this lesson.</i> Source: iicba  Search ARES for similar readings	Learners return to the initial model they began in Lesson 1 and revised afterwards (the bucket, the well, energy forms). Individually they add a new layer: (1) a labelled arrow for the woman's upward force of 200 N and the distance 8 m, with '$W = +1600 \text{ J}$'; (2) an arrow for gravity with '$W = -1600 \text{ J}$'; (3) a note 'net work = 0 J, so speed constant, KE unchanged'; (4) a link 'work done by the woman = energy transferred to the bucket = PE gain 1600 J'; (5) a small inset: lowering the bucket, signs reversed. They also leave a question mark box beside the pulley: 'Does the pulley change the work? Need Lesson 6 evidence.' They add a note on the energy source: chemical energy from ugali and githeri in the woman's body. They share with a partner and make one improvement from feedback.	Say: 'Your model must now show how energy gets into the bucket. Add force, distance and the sign of the work.' Pass out the model sheets from the earlier lessons. Walk around and ask: 'Where on your model can I see the force? The distance? The sign?' WAIT TIME: 10 seconds. Prompt: 'How does your model show where the 1600 J comes from?' (chemical energy from food in her body). Cold-call 3 learners to describe one change they made. Say: 'Keep the pulley as an open question: do not guess the numbers yet.' Collect the models for review. Close by asking a final question: 'If the woman did exactly 1600 J of work on the bucket by hand, what do you predict about Mwende using the pulley system?' WAIT TIME: 15 seconds, write the predictions on the board for use at the start of Lesson 6.	Cumulative model revision: adding force, distance, sign and energy-source labels to the same model across lessons shows how new evidence refines earlier ideas. Leaving an explicit 'unknown' box about the pulley models scientific practice, where gaps guide the next investigation.	Observable indicators: models include $W = Fd$ with 200 N and 8 m = 1600 J; correct signs (+1600 J, -1600 J, net 0 J); a link between work and the PE gain; a food (chemical) energy source; an open question about the pulley. Criteria: a learner who only draws the bucket with no arrows has not yet linked work with force and motion; give a prompt card with 'Where is the force? Which way does it move?'

D. TEACHER REFLECTION

After the lesson, check: Did learners separate everyday 'effort' from physics 'work'? Which groups still think holding a load is work? Did they understand that negative work is about direction, not 'less' or 'wasted'? Were the numbers consistent: $200\text{ N} \times 8\text{ m} = 1600\text{ J}$, -1600 J for gravity, net 0 J , and signs reversed for lowering? Did anyone start to guess the pulley numbers? Note groups needing extra practice with units and sign diagrams. Prepare Lesson 6 by collecting the DQB questions about the pulley, selecting 3 learner predictions about Mwende's effort, and checking that the rope and pulley materials or diagrams are ready.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a 2 kg load lifted 1 m needed about 20 N of force, giving $F \times d = 20\text{ J}$, and that holding the load still or carrying it sideways involved no distance in the direction of the force. For the bucket in the Kitui well, $200\text{ N} \times 2\text{ m}$, 4 m , 6 m and 8 m gave 400 J , 800 J , 1200 J and 1600 J , matching the PE table.
What did I learn?	We learned that work is done when a force moves an object in the direction of the force, $W = Fd$, measured in joules. Work is positive when the force and motion are in the same direction and negative when they are opposite. Mama Kavata does $+1600\text{ J}$ lifting the bucket, gravity does -1600 J , and the net work is 0 J at steady speed. When lowering the bucket, the signs are reversed.
How does this explain the phenomenon?	The direct lift needs 200 N over 8 m , so the woman does 1600 J of work, which becomes the 1600 J of PE gained by the bucket. The energy comes from the chemical energy in her body from food such as ugali and githeri. We still need to find out how the pulley system changes the force and distance and whether it changes the work, which we will investigate in Lesson 6.

LESSON 6 (80 minutes (2 x 40-minute periods)): Work and Energy: Hand Lift versus Pulley System at the Kitui Well

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners connect work done to change in energy, compare the work done by hand and with the pulley system when lifting the 20 kg bucket 8 m at Mama Kavata's well in Kitui, and refine the Driving Question Board (DQB) with new questions.
Knowledge	Work done on an object equals the change in its energy. The bucket gains 1600 J of PE ($200 \text{ N} \times 8 \text{ m}$) whether lifted by hand or with the pulley system. By hand: $200 \text{ N} \times 8 \text{ m} = 1600 \text{ J}$ of work. With the pulley system: effort $125 \text{ N} \times 16 \text{ m}$ rope = 2000 J of work input; 1600 J becomes PE and 400 J (20%) is transferred as heat (and a little sound) by friction in the pulleys and rope. The pulley gives a smaller force over a longer distance but does not reduce the energy needed.
Skills	Predicting, calculating work using $W = Fd$, constructing an energy-transfer table and Sankey-style diagram, analysing and interpreting data, using mathematical thinking (NGSS SEP: Using Mathematics and Computational Thinking; Constructing Explanations; Developing and Using Models), collaborating in groups, and asking questions.
Attitudes	Curiosity about why the pulley feels easier, honesty in reporting measurements, appreciation of the role of machines in rural life in Kitui County, and respect for others' ideas during group discussion.
Key Inquiry Question	1. How do we use energy in our day-to-day life? 2. How do machines make work easier?
Purpose in Storyline	Lessons 1-4 established that energy is stored as PE and KE, is conserved, and is partly transferred to heat and sound in real systems. Lesson 5 defined work as $W = Fd$ with positive and negative signs. Lesson 6 joins these ideas: work equals energy change, and the pulley system trades a smaller force (125 N) for a longer distance (16 m) while 400 J is transferred as heat. This answers the 'why easier, where does the energy come from, is any lost' puzzle and sets up Lesson 7 (friction and power) and Lesson 8 (MA, VR, efficiency).
Safety Notes	Use small masses (1 kg or 2 kg) with string and pulleys fixed to a strong frame or desk clamp; keep feet clear of hanging masses; do not pull the string suddenly; wear no loose items that may catch in pulleys; secure clamps before starting; if no pulley is available, use a smooth round pole or rod as a demonstration substitute under teacher supervision.

B. LESSON OVERVIEW

Mr Otieno's class in Kisumu returns to Mama Kavata's well in Kitui. Learners predict whether the pulley system lets Mwende do less work than a direct lift. They then gather evidence from a model lift (spring balance, string, two pulleys) and from the Kitui data: a hand lift needs 200 N over 8 m = 1600 J, while the pulley system needs 125 N over 16 m = 2000 J. Learners link work to energy change (the bucket gains 1600 J of PE either way), find that 400 J goes to heat in the pulleys and rope, and trace the original source to chemical energy from ugali and githeri. They finish by adding new questions to the DQB and revising their model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Thermal energy	Learners look at a photo or sketch of Mama Kavata lifting the 20 kg bucket (200 N) 8 m by hand and of	Display the sketch and say: 'Mama Kavata can barely lift the bucket, but 10-year-old Mwende lifts it with	Prediction with explicit reasoning (Predict-Observe-Explain): committing to an idea and a bar-	Check notebooks and listen to pairs for: (1) predictions that state a reason, not just a guess; (2)

from friction Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Conservation of Energy (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Mwende pulling the rope over the pulley system. Individually, then in pairs, they predict in their notebooks: (a) Does the bucket gain the same energy both ways? (b) Is the work Mwende does more than, less than, or equal to the work Mama Kavata does? (c) Where might the extra energy go? They sketch a quick bar chart for 'energy given by the person' and 'energy gained by the bucket' for each method.	the pulley system. Before we calculate anything, write your prediction.' Give 2 minutes of silent writing. Then say: 'Turn to your partner and compare your bar charts.' Cold-call 4 pairs for their prediction without judging. Record the three prediction options on the board with tallies. Ask: 'Using what we learned in Lesson 5, how would we calculate the work Mama Kavata does?' WAIT TIME 10-15 seconds before taking answers. Say: 'Keep your prediction; we will test it today.'	chart sketch before seeing data surfaces preconceptions (such as 'a pulley gives free energy' or 'less effort means less work') so learners can compare them to the evidence later and see their own thinking change.	whether learners draw equal bucket-energy bars for both methods; (3) whether learners suggest the extra energy goes to heat or friction. Tally the misconception 'the pulley reduces the work needed' and the idea 'less effort means less energy'. Aim to see at least 8 learners referring to Lesson 3-4 ideas of energy not disappearing.
Observe Phase VIDEO: Thermal energy from friction Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Conservation of Energy Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Part A (model lift, groups of 4): Learners hang a 2 kg mass (weight 20 N) from a spring balance and lift it 0.4 m straight up at steady speed, recording force and distance. They then set up a two-pulley arrangement (one fixed, one movable, 2 rope sections supporting the mass), pull the string with the spring balance, and record the effort and how far the string moved when the mass rises 0.4 m. Results are entered in a table: force, distance, work ($F \times d$). Part B (Kitui data): Learners compute from the Kitui data. Hand lift: $200\text{ N} \times 8\text{ m} = 1600\text{ J}$. Pulley system: $125\text{ N} \times 16\text{ m} = 2000\text{ J}$. Bucket PE gained: $200\text{ N} \times 8\text{ m} = 1600\text{ J}$. They calculate the difference, $2000\text{ J} - 1600\text{ J} = 400\text{ J}$, and compare with their classroom measurements. They also note the bucket speed is steady, so KE does not change and the whole energy gain is PE.	Circulate and say: 'What is your force reading? How far did your hand move compared with how far the mass rose?' Prompt: 'Which is bigger, the pulley effort or the weight? Which distance is bigger?' If groups get an ideal-looking result, say: 'Rub the pulley axle and rope; is the effort exactly half the weight?' Remind groups to keep the lifting speed steady and read the balance while moving. For Part B, post the Kitui data and ask: 'Where did the work number for Mwende come from?' WAIT TIME 10-15 seconds after each question. Cold-call 3 groups to report their effort-distance pairs and write them on the board. Ask: 'Is the pulley work equal to the bucket energy gain? What is the difference?'	Data table comparison and pattern finding: placing force, distance and work for both methods side by side lets learners see the pattern that force drops while distance rises and that work input with the pulley exceeds the energy gained. Real classroom data with friction makes the 'extra' energy visible.	Observe whether learners (1) read the spring balance at steady speed, (2) measure the string distance correctly, (3) calculate work with correct units (J), and (4) notice that the pulley effort is smaller but the distance is larger. Look for correct values: 1600 J, 2000 J, 400 J. Ask each group to show their completed table; at least 80% of groups should have correct work calculations and a statement that pulley work input is greater than the PE gained.
Explain Phase VIDEO: Work/energy problem with	Learners use the evidence to build a written explanation with a Claim-Evidence-Reasoning structure to answer: 'Why does the pulley	Say: 'Write a claim, then support it with numbers from our tables, then use the idea that work equals change in energy to reason.'	Claim-Evidence-Reasoning with an energy-flow diagram: requiring a claim backed by numbers and a scientific principle (work-energy,	Use a 4-point checklist on written CERS: (1) states that the bucket gains 1600 J either way; (2) states smaller force 125 N over longer

friction <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Conservation of Energy and Momentum (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	system feel easier for Mwende even though the bucket gains the same energy?' They apply the work-energy idea (work done on the bucket = change in its energy = 1600 J) and draw an energy-flow (Sankey-style) diagram: 2000 J from chemical energy in Mwende's body (from ugali and githeri) -> 1600 J PE of the bucket + 400 J heat (and a little sound) from friction in the pulleys and rope. They compare it with the direct lift, where 1600 J is the work needed with force 200 N over 8 m. They also name Lesson 4's idea that 'lost' energy means transferred to the surroundings, not destroyed, and note that 400 J is 20% of the input.	Model the first sentence stem: 'The pulley system makes lifting easier because ...'. Ask: 'Mwende puts in 2000 J. The bucket gains 1600 J. Where is the other 400 J?' WAIT TIME 10-15 seconds. Cold-call 4 learners, asking the next learner to add to or challenge the previous answer. Press: 'Is energy destroyed? Which Lesson 3-4 idea tells us?' Address the misconception directly: 'Does the pulley give extra energy or reduce the work needed?' (No: it gives a smaller force over a longer distance.) Ask: 'Where did Mwende's energy come from originally?' Have learners share Sankey diagrams in a gallery walk of 5 minutes.	conservation of energy) moves learners from describing to explaining. The Sankey diagram visualises transfer and shows that the energy is accounted for: $1600\text{ J} + 400\text{ J} = 2000\text{ J}$.	distance 16 m; (3) accounts for 400 J as heat/sound by friction; (4) states energy is conserved and not destroyed. Exit check: 'Write one sentence: why is work input greater than the PE gained?' Target: at least 75% of learners meet 3 of 4 criteria. Note learners who still say the pulley 'reduces the work' for reteaching.
Driving Question Board (DQB) Creation  VIDEO: Work/energy problem with friction - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili) Search ARES for similar videos  HTML: Conservation of Energy in Chemical Reactions (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES	Learners revisit the class DQB begun in Lesson 1. In groups, they sort existing questions into 'we can now answer' and 'still unanswered', and write the answers with evidence on sticky notes (for example, 'Same 1600 J of PE, but 125 N over 16 m instead of 200 N over 8 m'). They then write at least two new questions raised by today's work, such as: 'How fast is the work done, and does Mwende need more power?', 'How can we calculate how good the pulley system is?', 'How does friction between the rope and the pulley cause the 400 J of heat?', 'Would a ramp or lever also trade force for distance?' They place the new questions on the board, grouped by theme.	Say: 'Go to the DQB. Which questions have we answered today? Show me the evidence.' Give groups 5 minutes. Cold-call 3 groups to read an answered question and its evidence. Then say: 'What new question did the pulley data make you wonder about?' WAIT TIME 10-15 seconds. Highlight questions that point to Lesson 7 (friction and power) and Lesson 8 (MA, VR, efficiency): 'We will test these next.' Ask: 'Can anyone link a question to the 400 J of heat?'	Driving Question Board tracking: moving questions from 'unanswered' to 'answered' with evidence makes growth in understanding visible and gives learners ownership of what to investigate next, keeping the Kitui phenomenon central.	Check that at least 80% of groups provide an evidence-backed answer and at least 2 new questions each. Look for questions that reflect today's learning (friction losses, power, efficiency) rather than repeating earlier ones. Record which questions need Lessons 7 and 8.

for similar readings				
Model Building Phase  VIDEO: Work and the work-energy principle <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Conservation of Energy in Chemical Reactions Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to their Lesson 1 initial model of the bucket in the well and revise it. On a poster, they draw the well (0 m at the water surface, 8 m at the ground), the bucket, Mama Kavata (hand lift) and Mwende with the pulley system. They label: weight 200 N, PE gained 1600 J, effort 200 N over 8 m (1600 J) versus 125 N over 16 m (2000 J), energy source (chemical energy in food such as ugali and githeri), and 400 J transferred as heat by friction. They add arrows showing energy transfer and a short statement: 'Work done on the bucket = change in its energy.' Pairs explain their revised model to another pair and note what they changed since Lesson 1.	Say: 'Compare your model with Lesson 1. What do you now add, remove or change?' Provide a checklist: energy source, energy gained by bucket, energy input, energy wasted, force and distance. Circulate and ask: 'Where does the 400 J go in your model? Is the total energy the same before and after?' WAIT TIME 10-15 seconds. Cold-call 3 pairs to share one change they made and why. End with: 'Tomorrow we will find out how fast the work is done and how friction does negative work.' Collect models.	Cumulative model revision: updating the same model from Lesson 1 with quantitative evidence (1600 J, 2000 J, 400 J) shows how the explanation has grown from qualitative to quantitative and connects work, energy and conservation.	Rubric on models: (1) bucket energy gain of 1600 J shown for both methods; (2) pulley force 125 N and rope distance 16 m labelled; (3) 2000 J input, 1600 J useful, 400 J heat shown with arrows; (4) chemical energy from food shown as the source. Expect at least 70% to include all four. Learners who omit the heat transfer should be given a prompt in the next lesson.

D. TEACHER REFLECTION

Which learners still believe the pulley 'gives' energy or lowers the work needed, and what evidence changed (or did not change) their minds? Did the classroom model lift produce friction losses that learners could connect to the 400 J in the Kitui data? Were learners able to use $W = Fd$ with correct signs and units from Lesson 5? How well did the Sankey diagram help them see that $2000\text{ J} = 1600\text{ J} + 400\text{ J}$, and what scaffolds are needed for learners who struggled with the longer rope distance of 16 m? Which DQB questions about power and efficiency should be prioritised in Lessons 7 and 8?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that lifting the 20 kg bucket 8 m by hand needs 200 N over 8 m (1600 J), while with the pulley system the effort is only 125 N but the rope is pulled 16 m, giving 2000 J of work input. In our model lift, the pulley effort was smaller than the weight but the string moved further.
What did I learn?	We learned that work done on an object equals its change in energy, so the bucket gains 1600 J of PE either way. The pulley does not reduce the work needed; it trades a smaller force for a longer distance. The extra 400 J (20%) is transferred as heat (and a little

	sound) by friction, so energy is conserved.
How does this explain the phenomenon?	The pulley system feels easier for Mwende because she pulls with 125 N instead of 200 N, but she pulls 16 m instead of 8 m. Her body supplies 2000 J from chemical energy in food such as ugali and githeri; 1600 J becomes PE of the bucket and 400 J becomes heat in the pulleys and rope, so no energy is destroyed.

LESSON 7 (80 minutes (2 x 40-minute periods)): Work Against Friction and Power: Dragging the Maize Sack and Lifting Water

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence that friction does negative work that must be overcome (work = friction x distance) and that power $P = W/t$ measures how fast work is done, so they can explain why the pulley system needs more input energy than the PE gained and compare the power of hand lifting and pulley lifting.
Knowledge	Friction opposes motion, so the work needed to drag an object at steady speed equals friction x distance, and this energy is transferred to the surroundings as heat. Power is the rate of doing work, $P = W/t$, measured in watts (1 W = 1 J/s). Two jobs with the same work can need different power if done in different times.
Skills	Planning and carrying out an investigation (NGSS SEP 3): measure force, distance and time while dragging a sack. Analysing and interpreting data (SEP 4). Using mathematics (SEP 5): calculate $W = Fd$ and $P = W/t$. Constructing explanations (SEP 6). Developing and using models (SEP 2).
Attitudes	Curiosity about hidden energy transfers; teamwork and responsibility when sharing roles in the investigation; respect for others' ideas; appreciation of the effort of rural water-drawing families such as Mama Kavata's.
Key Inquiry Question	1. How do we use energy in our day-to-day life? 2. How do machines make work easier?
Purpose in Storyline	Lesson 5 taught $W = Fd$ and the signs of work. Lesson 6 showed that the pulley system needs 2000 J input for 1600 J of PE, with the extra 400 J going to heat. Lesson 7 gives learners direct evidence of how friction takes energy as heat and introduces power, so they can compare 80 W (hand) with 100 W input (pulley). This prepares Lesson 8, where MA, VR and efficiency are calculated.
Safety Notes	Use sacks filled with sand or maize of no more than 5 kg in class (the 50 kg sack is a scaled data example). Keep floors clear and dry and wear closed shoes. Learners drag in straight lines with a spring balance and do not run on stairs. For the stair activity, only learners who are healthy and willing take part, and the walk and run are done carefully using a short step set of 3 m (or data from the video). Others act as timers and recorders.

B. LESSON OVERVIEW

Mr Otieno's class in Kisumu returns to the puzzle from Kitui: the pulley system needs 2000 J of work input for only 1600 J of energy gained by the bucket. Where does the extra 400 J go? Learners first predict whether dragging a sack of maize across rough dirt or smooth concrete needs the same work. They gather evidence from a friction demonstration (scaled-down sack with a spring balance, and data for the 50 kg homestead sack: rough 150 N x 10 m = 1500 J in 25 s = 60 W; smooth 100 N x 10 m = 1000 J in 20 s = 50 W). They then define and calculate power with stair climbing (1500 J in 6 s = 250 W; in 3 s = 500 W) and with the well (hand 80 W; pulley input 100 W, useful output 80 W). They add friction as heat and power to the DQB and revise their class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall Mama Kavata's bucket and the 400 J puzzle from	Say: "The sack stays on the ground. It does not go up. So	Predict-Share-Revisit: committing to a written prediction exposes	Observe whether learners (1) give a reason with their prediction, not

 VIDEO: Work/energy problem with friction - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Friction (Flexbook) Source: CK-12  Search ARES for similar readings	Lesson 6. Mr Otieno shows a picture of the 50 kg sack of maize in the homestead store. Individually, in their notebooks, they predict and sketch: (a) is it harder to drag the sack 10 m across the rough dirt yard or the smooth concrete floor, (b) what happens to the energy used to drag it, because the sack does not gain height, and (c) if two people drag the same sack the same distance, one in 25 s and one in 50 s, what is the same and what is different? They write a prediction with a reason and share with a partner.	where does your effort go?" Display the picture and read the three prompts. Give 2 minutes of silent writing. Then WAIT TIME of 10 seconds after asking: "Which surface needs more work, and why do you think so?" Cold-call 4 learners (not volunteers) and record predictions in two columns on the board: 'Rough needs more' and 'Same work'. Ask: "Mwende's pulley needed 400 J extra. Could friction be involved?" Do not confirm yet; say: "Today we collect evidence to test these ideas."	prior ideas (for example 'no energy is used if the sack does not gain height'). Learners return to these predictions after the evidence so they can see their own thinking change. This links directly to the Kitui puzzle of energy that seems to disappear.	just a choice, (2) mention friction or the ground rubbing, (3) link the sack to the 400 J loss in the pulley. Tally on a clipboard how many predict that rough needs more work and how many say the sack needs no energy. Note learners who think work is only done when something is lifted, to be revisited in the Explain phase.
Observe Phase  VIDEO: Work/energy problem with friction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Friction Practice Source: CK-12  Search ARES for similar readings	Part A (about 25 min), friction investigation in groups of 4: roles are puller, spring-balance reader, distance measurer and timer/recorder. Groups drag a sand-filled sack (or school bag with 5 kg of sand) 5 m at steady speed on a rough surface (school yard or sacking cloth) and on a smooth surface (classroom floor), reading the spring balance (force) and recording distance and time. They compute $W = Fd$ for each surface. Then they study the homestead data for the 50 kg sack dragged 10 m: rough dirt friction 150 N, work 1500 J, 25 s; smooth concrete friction 100 N, work 1000 J, 20 s, and note that the sack's weight (500 N) is the same on both surfaces. They notice that the sack does not gain PE or KE (steady speed, no height change), yet work is done, and the ground and sack feel warm after rubbing (rub hands together or feel the sack base). Part B (about 25 min), power: learners watch or perform a safe stair-climb (20 steps of 0.15 m = 3 m for a 50 kg learner) with	Before the activity say: "Pull slowly and steadily. If the balance reading keeps changing, repeat the trial." Circulate and ask each group: "Is the force you read equal to friction? How do you know?" (steady speed means balanced forces). Cold-call 3 groups to report their rough and smooth force readings and ask: "Which is larger, and by how much?" WAIT TIME of 10 seconds. Show the homestead data table on the board and ask: "The sack weighs 500 N on both floors. Why is the work different?" Take answers from 3 learners. Ask: "Where did the 1500 J go if the sack did not rise?" For Part B, say: "Both walkers and runners do 1500 J. What is different?" Wait 10-15 seconds, then cold-call 3 learners. Introduce $P = W/t$ and $1 W = 1 J/s$ with the walking data ($1500/6 = 250 W$) and the running data ($1500/3 = 500 W$). Then ask learners to compute the well powers: hand $1600/20 = 80 W$; pulley input $2000/20 = 100 W$; pulley useful output $1600/20 = 80 W$. Ask: "Why is the pulley input	Data table analysis with a Claim-Evidence-Reasoning frame: learners complete a table of F, d, W, t, P across the sack, stair and well cases, and look for patterns (rougher surface, larger friction force, more work; same work in less time, more power). Comparing three different contexts in a single table lets them see that the same two ideas, $W = Fd$ and $P = W/t$, apply to a sack, a person and a well, and builds the concept of friction as an energy transfer to heat.	Check each group's table: Are units correct (N, m, J, s, W)? Does the learner use $W = F \times d$ with friction as F for dragging? Do they distinguish work from power (work is energy transferred; power is how fast)? Ask 3 learners individually: "What does 250 W mean?" Expected: 250 J of work done every second. Look for the misconception that more power means more work (the walker and runner do the same 1500 J). Listen for learners linking 'friction' with 'heat' without being told.

	timing: walking 6 s and running 3 s. They record the data in a table of work, time and power. They then use the well data: hand lift 1600 J in 20 s, and pulley input 2000 J in 20 s with useful output 1600 J in 20 s.	power higher than the hand lift power even though Mwende pulls less force?"		
Explain Phase  VIDEO: Work/energy problem with friction - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Friction (Flexbook) Source: CK-12  Search ARES for similar readings	Learners work in their groups to build a written Claim-Evidence-Reasoning explanation to answer: 'Why does the pulley system need 2000 J of input to give the bucket 1600 J of PE, and what does power tell us about Mama Kavata and Mwende lifting the bucket?' They use evidence from the sack (friction work becomes heat), the stair climb (power compares speed of work) and the well (80 W hand, 100 W input and 80 W output for the pulley). They also solve two short problems: (1) a 50 kg sack dragged 10 m on a rough floor with friction 150 N in 25 s: work and power. (2) Mama Kavata fills a second 20 kg bucket from the 8 m deep well in 40 s instead of 20 s: power ($1600/40 = 40$ W) and comment on how it compares with 80 W. They then revisit their Predict-phase ideas and correct them.	Say: "Write your claim first, then at least two pieces of evidence with numbers, then your reasoning in your own words." Give 8 minutes. Circulate and ask: "Which number shows the energy that went to heat?" and "Is the work done against friction positive or negative work by friction?" (friction does negative work on the sack; the person does positive work equal to $f \times d$). Cold-call 4 groups to share. After each, ask a different group: "Do you agree? What evidence would you add?" Use WAIT TIME of 10-15 seconds. Clarify, if needed, that no energy is destroyed: the 400 J in the pulley and the 1500 J in dragging are transferred as heat to the surroundings (consistent with Lesson 6). Do not teach efficiency yet: say, "In the next lesson we will find a number that tells us how good the pulley system is." Revisit the Predict board and ask learners to move or correct statements.	Claim-Evidence-Reasoning with peer critique: writing the reasoning chain (friction opposes motion, so effort must do work against it, the energy becomes heat, power tells how quickly it is done) forces learners to connect the mechanism with numerical data. Peer critique by another group catches the common confusion between work and power and strengthens argumentation (SEP 7).	Use a 3-level rubric: Level 3 states friction work converts input energy to heat with correct numbers (1500 J, 400 J) and gives $P = W/t$ calculations correct with units; Level 2 gets calculations right but explanation vague; Level 1 confuses work and power or says energy is lost/destroyed. Check problem answers: 1500 J and 60 W; 40 W. Quick exit question: 'A runner and a walker climb the same stairs. Who does more work? Who has more power?' Expected: same work, runner more power.
Driving Question Board (DQB) Creation  VIDEO: Work/energy problem with friction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners return to the class DQB started in Lesson 1. In groups they place sticky notes for (a) questions now answered by today's evidence, for example 'Where does the energy go when we drag something along the ground?' and 'Why does the pulley system need more energy input than the bucket gains?', with a short answer and evidence; (b) new questions, for example 'How can we measure	Say: "Take one sticky note that you can now answer. Write the answer with a number from today." Give 4 minutes. Cold-call 3 groups to read a note aloud and place it on the board; ask the class, "Does the evidence support that answer?" WAIT TIME of 10 seconds. Point to unanswered questions and ask: "Which tool could help us answer how efficient the pulley system is?" Highlight the	Driving Question Board sorting: categorising questions as answered, partly answered or new makes learning progress visible and keeps the investigation learner-driven. Writing answers with evidence reinforces that claims need data, and new questions become the purpose for the next lesson.	Look at the quality of sticky notes: do answers include numbers and units (for example 400 J as heat, 60 W)? Are new questions testable rather than yes/no? Record which groups connect friction to the pulley loss and which connect power to the 80 W and 100 W. Note learners who do not yet link the sack activity to the well and give them a targeted prompt: 'Where is friction in the pulley

 EXERCISE: Friction Practice Source: CK-12  Search ARES for similar readings	how good a pulley is?' 'Is the pulley system more or less powerful than lifting by hand?' 'Would oiling the pulley reduce the 400 J?'; and (c) a link to the main driving question about how pulleys reduce effort and how energy is transferred. They sort questions into: answered, partly answered, to investigate in Lesson 8.	question that leads to Lesson 8 (efficiency, MA and VR) and circle it. Ask 2 learners to explain how friction connects the sack and the pulley.		system?'
Model Building Phase  VIDEO: Work as the transfer of energy <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Heat, Temperature, and Thermal Energy Transfer (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners revise their class and individual models of the Kitui bucket system. On a large drawing of the well, pulley system, Mama Kavata, Mwende and the bucket, they add: (1) energy input from chemical energy in ugali and githeri; (2) friction in the pulleys and rope as a force opposing motion; (3) arrows for energy transfer: 2000 J in, 1600 J to PE of the bucket, 400 J to heat by friction; (4) power labels: hand lift 80 W; pulley input 100 W, useful output 80 W; (5) a small inset showing the maize sack drag with 1500 J as heat on rough ground. They write a short caption explaining why a smaller force of 125 N still needs more total work than the 1600 J gained and what power tells us about how fast the energy is delivered. They mark parts of the model that are still uncertain for Lesson 8, such as how to measure how well the pulley system works.	Say: "Add at least three new parts to your model today. Label every arrow with a number and unit." Give 10 minutes. Circulate and ask: "Where does the 400 J go? What form is it in?" and "What is the difference between the 100 W and the 80 W on your model?" (input power versus useful output power). Cold-call 3 learners to explain one addition to their model. WAIT TIME of 10-15 seconds before calling. Pair groups for a gallery walk of 5 minutes, with each pair giving one 'glow' and one 'grow' comment. Close by saying: "Next lesson we will turn these ideas into numbers that describe pulleys, levers and ramps."	Cumulative model revision with gallery walk: adding friction and power to a visual model of the phenomenon integrates energy conservation, work and rates into one explanation. Labelling every arrow with numbers makes the model testable, and peer feedback helps learners spot missing parts (SEP 2).	Checklist for each model: (1) 2000 J, 1600 J and 400 J shown correctly; (2) friction shown as the cause of heat; (3) chemical energy from food as the source; (4) power values 80 W and 100 W labelled correctly; (5) no suggestion that energy is destroyed. Collect captions as exit tickets. Mark learners who still say 'the pulley gives free energy' or 'energy is used up' for follow-up in Lesson 8.

D. TEACHER REFLECTION

Did learners treat the drag of the sack as real work even though the sack did not rise? Which groups used friction and heat spontaneously to explain the 400 J in the pulley system? Did anyone confuse work with power, for example by saying the runner does more work than the walker? Were spring balance readings steady enough for useful data, and how could the procedure be improved (for example, more trials or a smoother pull)? Which learners participated least and how can roles be rotated in Lesson 8? What evidence will you need to see in Lesson 8 so that learners can explain MA, VR and efficiency without losing the link to energy conservation?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that dragging a sack of maize across rough ground needs a larger friction force (150 N) than on smooth concrete (100 N), so more work is done over the same 10 m (1500 J versus 1000 J). We also saw that the same stair climb (1500 J) takes 6 s when walking and 3 s when running, and that the pulley system needs 2000 J input in 20 s (100 W) while hand lifting needs 1600 J in 20 s (80 W).
What did I learn?	We learned that friction does negative work on a moving object, so a person must do work equal to friction x distance to keep it moving at steady speed, and that this energy is transferred as heat. We learned that power is the rate of doing work, $P = W/t$, measured in watts where $1\text{ W} = 1\text{ J/s}$, and that doing the same work faster needs more power.
How does this explain the phenomenon?	The pulley system needs 2000 J of input although the bucket gains only 1600 J of PE because friction in the pulleys and rope transfers 400 J to heat, just as friction turns the work of dragging the maize sack into heat. No energy is destroyed. The pulley lets Mwende use a smaller 125 N force over a longer 16 m rope pull, and its useful output power of 80 W equals the hand lift power while its input power is 100 W.

LESSON 8 (80 minutes (2 x 40-minute periods)): Simple Machines, MA, VR, Efficiency and the Final Model: Why the Pulley Makes the Bucket Feel Light

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To let learners calculate mechanical advantage, velocity ratio and efficiency for pulleys, levers and ramps, and use this evidence to write the Final Explanation of how the pulley system reduces effort at the Kitui well while energy is conserved.
Knowledge	Learners will: (a) calculate $MA = \text{load/effort}$, $VR = \text{effort distance/load distance}$ and $\text{efficiency} = (MA/VR) \times 100\%$ for the pulley system (1.6, 2, 80%), a ramp (2.5, 4, 62.5%) and a crowbar (3.2, 4, 80%); (b) explain that a single fixed pulley has $VR = 1$ and only changes the direction of the force; (c) explain that a machine never reduces the work needed but trades a smaller force for a longer distance; (d) explain that the 400 J difference between 2000 J input and 1600 J output is transferred as heat by friction, so energy is conserved.
Skills	Developing and using models (revising the Lesson 1 model into a final model); analysing and interpreting data (comparing three machines); using mathematics (MA, VR, efficiency); constructing explanations (writing the Final Explanation with claim, evidence and reasoning); engaging in argument from evidence during the gallery walk.
Attitudes	Curiosity and perseverance in resolving the puzzle of why the same energy feels easier to supply; honesty in reporting measured data including the energy that goes to heat; respect for others' ideas in peer critique; appreciation of simple machines in everyday Kenyan life (wells, wheelbarrows, lorries, crowbars).
Key Inquiry Question	1. How do we use energy in our day-to-day life? 2. How do machines make work easier?
Purpose in Storyline	This is the final lesson of the unit. Lessons 1-7 gave learners energy as stored (potential) or present due to motion (kinetic), the KE and PE formulas, conservation of energy, energy losses as heat and sound, work $W = Fd$ with signs, the comparison of hand and pulley work, friction work and power. Lesson 8 brings these together: learners use MA, VR and efficiency to answer the driving question, complete the DQB, and compare their final model with their Lesson 1 model.
Safety Notes	Planks and crowbars: keep ramp planks flat and secured, load only small safe masses (e.g. 2 kg) in class, keep fingers clear of pulleys and rope, wear no loose scarves near pulleys, lift with bent knees. Use spring balances hung from firm supports. Do not stand under any suspended load. Crowbar demonstrations should use a light load and be supervised by the teacher.

B. LESSON OVERVIEW

Mr Otieno's class in Kisumu returns to Mama Kavata Mutua's well in Kitui. The 20 kg bucket (200 N) gains 1600 J of PE when raised 8 m, by hand or by pulley system. Learners first predict which of three machines (pulley system, ramp, crowbar) is most efficient. They then test or analyse data for the pulley system, ramp and lever, calculating MA, VR and efficiency. In the Explain phase they write the Final Explanation: Mwendu pulls 125 N over 16 m (2000 J input) instead of 200 N over 8 m; 1600 J becomes PE and 400 J (20%) goes to heat by friction; energy is conserved. They complete the DQB and compare their Lesson 1 model with the final model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Introduction to mechanical advantage Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mechanical Advantage (Flexbook) Source: CK-12  Search ARES for similar readings	Learners (10 minutes) look at photos of three Kenyan machines: Mama Kavata's pulley system at the well, two youths loading a 50 kg sack of maize onto a lorry 1.0 m high using a 4 m plank, and a farmer using a crowbar to lift a heavy stone. Individually they write in their notebooks: (1) Which machine will need the smallest effort? (2) Will any machine reduce the work needed to raise the load? (3) Rank the three from most to least efficient and give a reason. They then share in pairs and record predictions on a class chart. Link to phenomenon: they also predict whether a single fixed pulley alone would make the bucket easier for Mama Kavata.	Display the three photos. Say: 'Mama Kavata's bucket gains 1600 J whether she lifts it by hand or with the pulley system. Today we find out why it still feels easier. Before I give you any numbers, predict.' Give 2 minutes of silent writing. Then say: 'Turn to your partner and compare rankings.' Cold-call 4 pairs: 'Which did you rank most efficient, and what is your reason?' WAIT TIME of 10 seconds after each question before calling on a learner. Record all predictions without judging. Ask: 'Does anyone think a machine can reduce the work needed? What would that mean for energy?'	Predict-Observe-Explain (POE) with a ranking prediction. Committing to a ranking and a reason exposes the misconception that machines create or save energy, so the later data create cognitive conflict and learners are motivated to revise their thinking.	Look for: learners who write that a machine 'gives extra energy' or 'reduces work' (misconception to address later); learners who mention friction as a reason for lower efficiency; learners who give a reason beyond 'it looks easier'. Tally on the board how many predict the ramp, the pulley system or the crowbar as most efficient. Target: at least 80% of learners write a ranking with a reason before the discussion.
Observe Phase  VIDEO: Introduction to mechanical advantage - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Mechanical Advantage Practice Source: CK-12  Search ARES for similar readings	In groups of 4 (30 minutes) learners rotate through three stations, using real equipment where available or the data cards where not. Station 1 Pulley system: using the rig (or the video/photo of Mama Kavata's well), record load 200 N, effort 125 N, load distance 8 m, effort distance 16 m; also observe that a single fixed pulley alone has VR = 1 and ideal effort 200 N, so it only changes direction (Mwende can pull down using her body weight). Station 2 Ramp: 50 kg sack (500 N) loaded 1.0 m high with a 4 m plank; effort 200 N. Station 3 Lever (crowbar): effort arm 1.2 m, load arm 0.3 m, load 400 N, effort 125 N. For each machine they complete a table: load, effort, MA = load/effort, VR = effort distance/load distance, work input, work output, efficiency = (MA/VR) x 100%. Expected results: pulley system MA 1.6, VR 2, 80% (input 2000 J, output 1600 J); ramp MA	Circulate. Say: 'Before you calculate, tell me what the load is and what the effort is for this machine.' Ask at the pulley station: 'How many rope sections support the bucket? How far must the rope move for the bucket to rise 8 m?' WAIT TIME of 10-15 seconds before accepting answers. At the single fixed pulley say: 'Mwende has a mass of 30 kg. Why can she pull down on the rope more easily than she can lift directly?' (She uses her body weight; the pulley only changes direction, VR = 1.) At the ramp ask: 'Work in is 800 J but work out is 500 J. Where did the 300 J go?' Cold-call 3 groups to report one MA and one efficiency each. Remind: 'Use g = 10 N/kg and write units. MA and VR have no unit.' Pause at 20 minutes and check that every group has all three rows filled.	Data table comparison (analysing and interpreting data). Putting three machines side by side in one table lets learners see the pattern: smaller effort always comes with a longer effort distance (MA up, VR up) and efficiency is always below 100% because of friction. Spotting the pattern across cases builds the general idea instead of one isolated fact.	Observable indicators: correct MA = $200/125 = 1.6$, VR = $16/8 = 2$ and efficiency = $1.6/2 = 80\%$ for the pulley system; correct ramp (2.5, 4, 62.5%) and crowbar (3.2, 4, 80%); learners notice that MA is less than VR in every row; learners can say that the single fixed pulley has VR = 1. Use a checklist: tick each group's three correct rows. Intervene if a group divides effort by load (inverted MA) or forgets to multiply by 100 for efficiency.

	2.5, VR 4, 62.5% (input $200 \times 4 = 800$ J, output $500 \times 1 = 500$ J); crowbar MA 3.2, VR 4, 80%. Finally learners compare with their predictions.			
Explain Phase  VIDEO: Mechanical advantage (part 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Power and Efficiency (Flexbook) Source: CK-12  Search ARES for similar readings	Learners (20 minutes) write the Final Explanation individually using Claim-Evidence-Reasoning, then critique a partner's work. Required elements: (1) The bucket gains 1600 J of PE ($200 \text{ N} \times 8 \text{ m}$) whether lifted by hand or by the pulley system. (2) By hand: effort 200 N over 8 m = 1600 J, power 80 W. With the pulley system: effort 125 N over 16 m = 2000 J input, so Mwende, a 10-year-old, can do it with a smaller force. (3) The pulley does not give extra energy or reduce the work needed; it trades a smaller force for a longer distance (VR = 2, MA = 1.6). (4) The energy comes from chemical energy in food such as ugali and githeri. (5) 1600 J becomes PE of the bucket and 400 J (20%) is transferred as heat (and a little sound) by friction in the pulleys and rope, so efficiency = $1600/2000 = 80\%$ and total energy is conserved, none destroyed. (6) Machine comparison: the ramp (62.5%) loses more energy than the crowbar and pulley system (80%). Then learners do a gallery walk of 2 minutes per poster to critique one other pair's explanation.	Say: 'Write as if explaining to Mama Kavata why Mwende can do what she could not. Use at least three numbers from today.' Provide a sentence frame on the board: 'The pulley system reduces the effort from ___ N to ___ N because the rope moves ___ m instead of ___ m, so the work input is ___ J, while the bucket gains ___ J. The difference of ___ J is ___.' Give 8 minutes of silent writing, then say: 'Swap with your partner and underline one number and one piece of energy reasoning.' Cold-call 5 learners to read their strongest sentence. WAIT TIME of 10-15 seconds before each call. Press with: 'If the bucket gains the same energy, why does it feel easier?' and 'Where did the 400 J go? Is it destroyed?' Correct any statement that energy is 'lost' by adding 'transferred to the surroundings as heat'. Challenge: 'Could an ideal pulley system with no friction have 100% efficiency? What would the effort be?' (100 N, MA = 2 = VR.)	Claim-Evidence-Reasoning (CER) with peer critique. Learners must tie the claim that energy is conserved to numerical evidence from the 80% pulley system, the ramp and the crowbar, and justify with the work-energy link and conservation of energy from earlier lessons. Peer critique makes weak reasoning visible and strengthens argumentation.	Use a 4-point rubric on the Final Explanation: (1) states 1600 J PE is equal for both methods; (2) states 125 N over 16 m = 2000 J input versus 200 N over 8 m = 1600 J; (3) states 400 J (20%) transferred as heat by friction, none destroyed; (4) uses MA 1.6, VR 2, 80% correctly. Target: 75% of learners score 3 or 4. Flag learners who say the pulley 'saves work' or 'makes energy' for a follow-up conference.
Driving Question Board (DQB) Creation  VIDEO: Mechanical advantage (part 3)	Learners (10 minutes) return to the class DQB started in Lesson 1 and sort every question into three groups: 'Answered with evidence', 'Partly answered', and 'New questions'. In groups they choose two questions, write a short evidence-based answer on a	Say: 'Our board began with Mama Kavata's tired arms. Let us see what we can now answer.' Read aloud three questions and ask: 'What evidence from this unit answers it?' WAIT TIME of 10-15 seconds. Cold-call 4 groups to place a sticky note and justify it.	Driving Question Board revisiting. Sorting questions by whether they are answered makes progress visible, links evidence to questions across the unit, and shows learners that science generates new questions, sustaining curiosity.	Check that each sticky note cites specific evidence (a number or a lesson result) rather than a bare statement. Observable target: every group moves at least 2 questions to 'Answered', and every learner writes one new question. Note questions that still show

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Power and Efficiency Practice Source: CK-12 Search ARES for similar readings	sticky note (e.g. 'Why does it feel easier? Because the effort is 125 N instead of 200 N over a longer rope, VR = 2'; 'Is energy lost? 400 J goes to heat by friction; energy is conserved'), and attach it next to the question. Each learner writes one new question that remains, such as 'How could we raise the efficiency above 80%?' or 'How does a bicycle gear work as a machine?'	Prompt: 'Which lesson gave us this evidence?' (use only: Lesson 4 energy losses, Lesson 6 hand vs pulley, Lesson 7 power.) Ask: 'Which questions are still open and how might we investigate them?' Point out that scientists keep new questions. Photograph the final DQB for the record.		confusion about work versus energy for re-teaching in revision.
Model Building Phase VIDEO: Mechanical advantage (part 2) -dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos HTML: Energy Efficiency (Flexbook) Source: CK-12 Search ARES for similar readings	Learners (15 minutes) take out their Lesson 1 initial model (qualitative, no formulas) and draw a final model on a new page of the Kitui well. The final model must show: the bucket at 0 m and 8 m with PE of 0 J and 1600 J; two energy-flow arrows for hand lift and pulley system (1600 J in versus 2000 J in); the rope pulled 16 m with a 125 N effort; the 400 J heat arrow leaving the pulleys and rope; the source of energy (chemical energy from ugali or githeri in Mama Kavata or Mwende); and labels for MA 1.6, VR 2 and efficiency 80%. Beside it, learners write a 3-sentence comparison: 'In my Lesson 1 model I thought _____. Now I know _____. The evidence that changed my mind was _____.'	Say: 'Place your first model beside your final one. What is the biggest change?' Give 10 minutes of drawing, then 5 minutes of sharing in pairs. Prompt: 'Does your model show where the energy comes from? Does it show where the 400 J goes?' and 'Can you show on your model why the force is smaller?' Cold-call 3 learners to explain one change from Lesson 1 to the final model. WAIT TIME of 10-15 seconds. Close by asking the driving question aloud: 'How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process?' and take 2 learner answers. Collect models for assessment.	Cumulative model revision with explicit comparison of first and final model. Seeing how the model grew from a qualitative idea to a quantitative, energy-conserving explanation lets learners recognise their own conceptual change and consolidates the unit.	Model checklist: includes 1600 J PE at 8 m; both energy routes (1600 J versus 2000 J input); 400 J heat arrow; chemical energy source; MA 1.6, VR 2, 80%; smaller force with longer distance. Target: at least 5 of 6 features in 80% of models. The comparison statement must name at least one changed idea and the evidence that changed it. Use model quality to give end-of-unit feedback.

D. TEACHER REFLECTION

Which learners still believe that a machine reduces the work or creates energy, and which piece of evidence (the 2000 J input versus 1600 J output) convinced or failed to convince them? Did learners use MA and VR to describe the force-distance trade, or did they only memorise the formulas? Did every group account for the 400 J as heat transferred to the surroundings rather than 'lost' or destroyed? How well did the final model improve on the Lesson 1 model, and which learners need further support with calculations or with explaining energy conservation? What local examples (wheelbarrows, bicycles, crowbars, lorry ramps) could deepen the next unit?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the pulley system needs an effort of 125 N for a 200 N load, with the rope pulled 16 m while the bucket rises 8 m. We observed that the ramp needs 200 N for a 500 N sack over 4 m and the crowbar needs 125 N for a 400 N load. In all three machines the work input was more than the useful work output.
What did I learn?	We learned that $MA = \text{load/effort}$, $VR = \text{effort distance/load distance}$ and $\text{efficiency} = (MA/VR) \times 100\%$. The pulley system has MA 1.6, VR 2 and 80% efficiency; the ramp has 2.5, 4 and 62.5%; the crowbar has 3.2, 4 and 80%. A single fixed pulley has VR = 1 and only changes the direction of the force. A machine never reduces the work needed; it trades a smaller force for a longer distance.
How does this explain the phenomenon?	The bucket gains 1600 J of PE whether Mama Kavata lifts it by hand or Mwende uses the pulley system. The pulley lets Mwende pull 125 N instead of 200 N, but over 16 m instead of 8 m, so 2000 J of chemical energy from food (ugali, githeri) is put in. 1600 J becomes PE of the bucket and 400 J (20%) is transferred as heat by friction in the pulleys and rope. Energy is conserved because none is created or destroyed.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.